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AI and ML empowering 5G and shaping the 6G future: Models, metrics, architectures, and applications

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ABSTRACT

Artificial Intelligence (AI) and Machine Learning (ML) technologies are becoming more important in wireless telecommunications networks, especially in the transition from 5G to 6G, a more advanced AI networking environment. While in 5G networks AI is used basically to get better performance from the individual tasks, in 6G, AI will be a model that is used at each layer of the system design-from the physical retransmission of the signals right through to the management of the services. The paper will examine the advanced AI technologies of Deep Learning, Reinforcement Learning, Generative Models, and Federated Learning, and their impact on core processes in the networking framework like beamforming, channel estimation, spectrum access, and anomaly detection which are evaluated against core metrics of accuracy, latency, power consumption, privacy, and comprehensibility. In the process of going beyond technical detail, the review situates AI-based wireless innovations in different fields including autonomous vehicles, telesurgery, industrial IoT, and smart cities. It also points out the persistent challenges, such as data scarcity, real-time inference, edge deployment, and ethical concerns, and presents some promising future research directions, including digital twins, AI–quantum convergence, and regulatory frameworks. This work presents a strategic roadmap to achieve scalable, secure, and intelligent 6G networks by providing a cross-layer and cross-domain synthesis.

1. Introduction

Since its inception as 1G in the 1980s, wireless communication has totally changed how people interact and live. 1G allowed basic mobile voice calls using analog modulation of electromagnetic waves. The next generation was 2G in the 1990s, which made voice digital and opened the door to consumer text messaging, taking real-time remote information exchange to the next level. In 2000, the world was introduced to 3G, which enabled the mobile device to function as an Internet device, allowing the use of early apps and Web browsing [1,2]. The next generation was 4G, released in 2010, which was many times faster than the earlier generations, enabling data transfer for excellent streaming of video, smooth video conferencing and real time enjoyment of the Internet. The latest acquisition by the telecoms is 5G, which is creating a period of hyper connection of tremendous speeds with little delay and the ability to connect people to not only other people but machines, transportation, and smart devices. One result has been a tremendous change in communication, work, and social life, making the world immeasurably connected while now having the ability to communicate through the fastest communications available. As we stand on the

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threshold of 6G, the paradigm is shifting from connected things to connected intelligence. 6G is not an evolutionary prerequisite of 5G, but a paradigm shift where AI and ML will be natively integrated into the fabric of the network technical architecture [3–5]. It envisages a scenario of semantic communication, digital twins, brain-computer interface, reconfigurable intelligent surfaces, and quantum-secured links that will necessitate intelligent, adaptive, and context-aware communication. The technological momentum towards 6G and the role of AI/ML in the transformation can only be appreciated by taking a look at the historical advancement of wireless generations [6–8]. Fig. 1 below outlines this evolution across key performance and application dimensions:

In contrast, 6G considers AI/ML not only as an intrinsic and ubiquitous layer but one which evolves in tandem with the communication system per se. Learning/reasoning and adaptation occur natively in the air interface, protocol stack, and orchestration of services, enabling the network to sense, recognize and act autonomously [9,10]. Thus, the transition from 5G to 6G represents a fundamental shift: AI in 5G enhances the network, whereas AI in 6G constitutes the network, driving a move from optimization to cognition, and from connected devices to connected intelligence. In 5G AI acts as a separate optimizer performing pre-defined tasks whereas, in 6G AI is a part of the architecture of the system being located across different layers of the system that enables learning, reasoning and real time adaptation thus making it a built-in feature rather than a utility that is add-on [11].

This is not just a move from 5G to 6G: it's a change in perspective towards the wireless network and how it's designed and deployed. While for 5G sometimes it would be enough to just deploy AI in order to predict, for example, traffic patterns, or detect anomalies, for 6G the goal is to redesign the AI so it actually becomes the CNS of the wireless network [12,13]. 5G networks utilize AI/ML primarily as optimization tools in an existing architecture, while in 6G, intelligence becomes a design component from the start. In 5G, AI supports certain operational functionalities such as traffic prediction, beam management, and energy optimisation; these operate over pre-defined control mechanisms.

Incorporating AI and ML across all levels from physical medium to application services means that predictive control, semantic communication, and real-time provision of adaptive resources can be realized. With the advent of technologies such as reconfigurable intelligent surfaces (RIS), THz communication, digital twins, and brain–computer interaction applications, decision-making needs to be fast, reliable, distributed, and explainable, which is best achieved via AI-driven mechanisms [14,15]. While 5G primarily leveraged AI for performance optimization and automation, the leap to 6G redefines the very objective of communication itself. The next generation is expected to optimize the manner in which data is transferred and also transform what is transferred and why. With its introduction will come an era when meaning, intent, and context become essential elements of network intelligence, rather than by-products of data transfer.

In addition to more improvements, the greatest expected change in 6G is the transition from data centric communication to meaning-centric communication (or semantic communication). Semantic communication is beyond the traditional networks' goals of the greatest fidelity in transmitting bits, to give emphasis to the attitudes and meaning in the context of the information being transmitted [16,17]. For example, in autonomous driving or telesurgery, the network can transmit only semantic essences-such as "lane clear" or "incision complete"-instead of sending each and every sensor frame or control signal. This drastically reduces bandwidth and latency by orders of magnitude while retaining decision accuracy intact. This is, in fact, a cognitive leap from throughput optimization via AI models capable of representation, reasoning, and contextual inference [11,18]. At the same time, AI-driven digital twins will offer real-time virtual twins of networks and services with the capacity provide closed-loop learning from which continuous prediction, optimization and fault recovery can take place. Together, semantic communications and digital twins embody the mega-trend of 6G, systems that not only connect but also understand and predict the world which they will serve. In this respect, AI presence in all layers of the wireless stack will not merely be a means of enhancement but a sine qua non for the realisation of this vision [19]. The intelligence encompassing semantic understanding and predictive digital twinning must be integrated at every level, from signal processing and resource management to service orchestration and user experience. It must ensure coherence across the meaning, action, and communication. This shift establishes the foundation for 6G as an AI-native ecosystem, where learning and reasoning are intrinsic network functions rather than external add-ons.

This importance of AI/ML is especially crucial considering the multi-dimensional complexity of next generation networks. Classical strategies based upon rules or determinism fail to scale under high user density or heterogeneous traffic patterns or when variable spectrum conditions are introduced [20,21]. Conversely, ML techniques provide model-free, data-driven techniques for such optimization such that the networks can learn via experience, adjust to contextual modification, and respond autonomously to unanticipated scenarios. Today, AI is being recognized

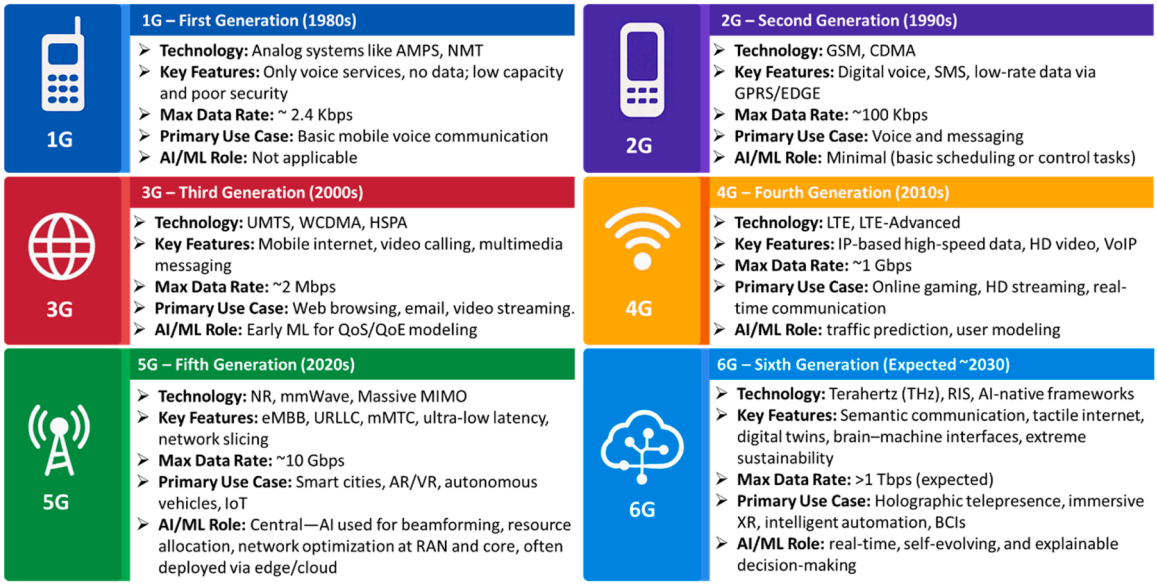


Fig. 1. Evolution of Mobile Communication Technologies from 1G to 6G.

as a necessary tool in many areas of wireless communication. For example, signals can be classified with the use of convolutional neural networks (CNNs), reinforcement learning (RL) is employed to enforce the smart access of the spectrum, federated learning allows for the training of networks at the edge without raw data needing to be transmitted, and generative adversarial networks (GANs) can be employed to produce a realistic model of the channel for testing. The reciprocality is illustrated here, in that, AI can render wireless networks smarter and operate at efficiencies not hitherto thought possible, while wireless networks can either provide the data or the environment for AI to grow and develop [9–23]. On one hand, AI enhances the reliability, performance, and scalability of networks. On the other hand, 5G and 6G networks serve as enablers for large-scale AI applications, such as collaborative robotics, real-time healthcare diagnostics, and autonomous mobility ecosystems. This co-evolution is central to building truly intelligent and immersive infrastructures [24–26].

This review seeks to provide a thorough and technically rigorous examination of the influences and impacts of AI and ML on existing 5G infrastructure as well as the future development of the 6G networks of tomorrow. A precise picture is being sought to clarify the contribution that the modulation of advanced AI impactors such as deep learning, reinforcement learning, graph neural networks, generative models and distributed learning could affect such vital functions of communications as beam-forming, resource-allocation, channel estimation and anomaly-detection, It will also discuss the conclusions that can be deduced on their effects in terms of deployment scenarios concerning essential measures such as inference time, energy-efficiency, privacy, expalainability and cost.

What sets this review apart from existing literature is its cross-layer, multidomain, and future-centric synthesis. While most papers focus either on AI models or isolated use cases, this paper spans several dimensions: technological, architectural, and application-driven, thus providing a 360-degree view of the AI-6G interface [27–30]. This review not only summarises the current developments but also the challenges of using AI in networks, the ethical difficulties, the failure of current specifications. The objective of this review was to provide researchers, industry, and policy makers with an understanding to develop and manage the wireless world of AI.

2. Pillars of 6G and the role of AI/ML

The move from 5G to 6G is motivated by more than just data rate improvements; it is motivated by a larger vision that unifies communication, computation, control, and sensing into a smart fabric. 6G stands on three crucial pillars (see Table 1) each of which will represent an evolution of current capabilities in 5G combined with AI and emergent technologies [31]. These include:

2.1. Immersive communication (eMBB evolution)

The evolution of enhanced Mobile Broadband (eMBB) is upping the ante for immersive holographic, extended reality, and multisensory communication. This requires unprecedented speeds, ultra-low latency, and tight synchronization. AI and ML enhance this with intelligent predictions of what a user will see by compressing only data that has a meaningful impact and allocating resources just when needed. Deep learning-based semantic encoders focus on transmitting essential information rather than raw pixels; much lower bandwidth usage can be achieved without compromising high-quality experiences. Recent works such as demonstrate AI-assisted holographic streaming using neural radiance fields, achieving significant latency reduction through intelligent content prioritization [32,33].

Table 1
Pillars of 6G and the Role of AI/ML [42–57].

6G Pillar	Evolution From 5G	AI/ML Contribution
Immersive Communication	eMBB → Holographic/XR	Semantic encoding, perceptual compression
Massive Communication	mMTC → Intelligent IoT	RL/Federated access optimization
HRLLC / URLLC	URLLC → Predictive Edge Control	RL-based latency prediction
Ubiquitous Connectivity	5G NTN → Global Fabric	GNN & FL-based topology control
AI-Native Communication	Optimization → Cognitive Control	Semantic Transformers, Digital Twins
Integrated Sensing and Communication (ISAC)	Radar & Data Fusion	AI-generated sensing parameters

2.2. Massive communication (mMTC evolution)

The 6G system naturalizes massive Machine Type Communications (mMTC) as a massive, intelligent and cooperative communication where tens of billions of devices across industrial, agricultural and urban ecosystems are connected through machine attachment and integrated AI technologies enable dynamic slot sharing, energy efficient access and context aware scheduling across heterogeneous IoT environments. Also using Federated and Reinforcement learning approaches for minimization of collision and congestion through distributed optimisation under non IID data conditions. For example, used Deep Q-Networks for adaptive random access, improving mMTC scalability and fairness under high load [34–37].

2.3. Hyper-Reliable and low-latency communication (HRLLC / URLLC evolution)

Mission-critical 6G applications like telesurgery, autonomous vehicles, and industrial automation require 99.99999 percent availability and sub-millisecond latency characteristics. AI-powered edge computing and network slicing are major enablers for helping achieve these goals. Reinforcement learning algorithms serve to dynamically allocate the resources of radio, compute, and transport, while predictive analytics will predict the state of a link before a performance degradation occurs. For example, applied model-based RL for URLLC resource prediction, achieving adaptive latency control in vehicular environments. Hybrid cloud–edge inference frameworks further reduce processing delays, enabling real-time feedback in critical applications [38–40].

2.4. Ubiquitous connectivity

6G aims to realize ubiquitous and pervasive network access for everyone and everything via terrestrial, aerial and satellite links as seamlessly woven into a global fabric. Ubiquitous connectivity is supported by Artificial Intelligence through topology-aware routing, channel predictions and self-healing control loops. Graph Neural Networks (GNN) and meta-learning ensure intelligent adaptation of non-terrestrial networks (NTN) and their dynamic topologies. A notable study by demonstrated AI-enhanced Low-Earth-Orbit (LEO) satellite scheduling using federated learning, reducing inter-satellite interference and improving coverage continuity [41–44].

2.5. AI-Native communication

AI is not a broadband optimization tool for 6G; it is one of the overarching design vertices that govern the semantics of communication, intent identification and cognitive orchestration. This vertex emphasizes the fusion of communication and cognition skills, where networks can perceive thought around communication, reason about it and learn from it. The salient constructs in this vertex are semantic communication, digital twins, and self-evolving protocol stacks. For instance, semantic-aware transformer networks have demonstrated bandwidth savings of up to 80 % compared with conventional bit-level

transmission by focusing on meaning rather than symbol fidelity [45].

Semantic communication represents a shift in the paradigm of communication from data-oriented to meaning-oriented communication. Rather than communicating signals with the utmost accuracy, the focus is on communicating just the task relevant information which maintains the semantics (or meaning) of the information. In applications such as autonomous vehicles or telesurgery, for example, the net may transmit the more succinct semantics of the language community, rather than the raw sensor streams, thus achieving better communication by vastly diminished bandwidth and reduced latency with preserved decision making fidelity [46,47]. Formally, this process can be expressed using a rate–distortion formulation as:

$$\min \mathrm{E}[d(\widehat{y}, y)] \mathrm{s.t.} \mathrm{E}[\ell(\mathrm{f}(x))] \leq \mathrm{R}$$

where fdenotes the semantic encoder mapping an input xinto a compact representation, g is the semantic decoder reconstructing the task-relevant output $\widehat{y}$, yis the ground-truth label or intended task output, $\mathrm{d}(\widehat{y}, \mathrm{y})$measures the semantic distortion (e.g., task loss or accuracy gap), and Ris the communication-rate constraint. This framework integrates task-oriented compression and transmission, ensuring that only information essential for successful inference is sent.

In the AI-native landscape of 6G, semantic communications will increasingly occur alongside digital twin-led optimization, predictive modelling, anomaly detection, and continuous improvement of policies as they flow from digital twin models of the physical entities being represented. Hence, networks will transmit not just contextual information, meaning, and context, but also the intention associated with context, which will allow for contextual-aware, energy efficient and goal-oriented information transfer [48,49]. Ultimately, the integration of AI across every layer of the wireless stack transforms networks from passive data conduits into cognitive ecosystems capable of perception, reasoning, and autonomous adaptation, fulfilling the vision of 6G as a learning and self-optimizing infrastructure.

2.6. Integrated sensing and communication (ISAC)

ISAC is one of the most transformative areas of 6G, achieving a combination of radar-like sensing and data communication into a single shared waveform. This epitomizes scenarios in which networks perceive the surroundings, detect objects, assess usable resources, and localize users while concurrently transmitting data. Furthermore, this captures the role of AI in learning how to create the optimal sensing parameters, manage interference between the tasks of sense and communication, and perform environmentally scanning signal processing [50,51]. For example, used deep generative models to determine passive target locations and optimize sensing parameters for multi-target tracking in ISAC-enabled 6G networks. Likewise, demonstrated reinforcement learning (RL)-based waveform design for joint communication and sensing that improved 25 % detection accuracy with spectral efficiency. Through AI-driven ISAC, 6G networks will evolve from data conveyors to environment-aware cognitive systems capable of situational understanding and adaptive decision-making [52,53].

State-of-the-art research is continuing to advance the performance of ISAC systems by improving both signal design and the interaction of smart surfaces with signals. It applies a technique called Mutual Information–based pilot design that enables wireless systems to estimate the communication channel and detect objects simultaneously, without any degradation in communication quality [54]. At the same time, engineers are working on multifunctional RIS that can do much more than reflect signals. Today, these reconfigurable intelligent surfaces are being designed to support sensing, communication, and even wireless power transfer on a single platform for making future 6G networks energy-efficient and more aware of their environment [55]. These developments together support AI-driven waveform and resource management, showing how both signal-level and surface-level designs can work together to achieve these dual goals on next-generation wireless networks.

2.7. AI-Driven enhancement of IMT-2030 key performance indicators

According to the International Telecommunication Union (ITU-R), IMT-2030 is the framework for the performance of 6G (as it goes beyond the eight KPI families of IMT-2020 to include also intelligence, sustainability and trustworthiness in addition to the basic metrics in the domain, the peak data rates, latency, reliability, spectral efficiency, energy efficiency and number of users in the area framework). AI is expected to act as a unifying accelerator for meeting, or surpassing, these indicators across all network layers (see Table 2).

3. AI/ML models and techniques for wireless communication

AI and ML have gone from helper-tool roles in 5G to the key enablers of the evolution to 6G-based networking systems and services that are adaptive, data-driven and intelligent. Whereas traditional networking approaches are dependent on hard-wired rules, AI/ML models can learn complex relationships from spatiotemporal distributions of data, compute network resource synergies in real-time mode and enhance system robustness and reliability in stochastic environments that can be unpredictable and dynamic [74]. In wireless networks, these types of learning approaches are key towards the realization of many critical elements of wireless networking including channel estimation, beam forming, spectrum allocation, mobility prediction and anomaly detection. Key models of interest include variations on deep learning architectures (CNNs, RNNs, LSTMs, GRUs), reinforcement learning for dynamic (real-time) decision making, generative models for synthetic channel generation and federated learning for exploration of edge devices under privacy-aware deployments. The choice of model depends on the task's spatial, temporal, and contextual complexity as well as deployment constraints like latency, interpretability, and compute availability [74,75].

3.1. Deep learning models

3.1.1. Convolutional neural networks (CNNs)

CNNs are particularly effective for spatial feature extraction and are commonly used for modulation classification, interference detection and CSI estimation in wireless communications as indicated in Fig. 2(A). CNNs can learn hierarchically derived concepts from convolutional filters and thus can extract frequency, time and spatial domain features from raw I/Q signal samples or even using spectrograms. Their ability to reduce dimensionality while preserving relevant patterns makes them ideal for physical-layer tasks such as beam selection in mm Wave systems or image-like data processing in massive MIMO setups [76–78].

3.1.2. Recurrent neural networks (RNNs)

RNNs are built to work with sequential data and are used in time series prediction problems in wireless systems, as shown in Fig. 2(B). They are preferably suited for mobility prediction, traffic load forecasting, and dynamic spectrum access, where current decisions depend to a large extent on the previous states of the network. However, due to vanishing gradient issues and difficulty in learning long-term dependencies, vanilla RNNs are often replaced by more advanced variants like LSTMs or GRUs in practical applications [79,80].

3.1.3. Long short-term memory networks (LSTMs)

LSTM architecture enhances RNNs with the capability of learning over long-term time scales by introducing memory cells and gating mechanisms to capture the temporal dependencies intrinsic in time-series applications. In wireless communication problems, LSTMs have proved useful in dynamic channel modelling, handover prediction, and user trajectories in ultra-dense networks. Their ability to maintain

Table 2
AI contributions toward achieving ITU-R IMT-2030 key performance indicators for 6G.

IMT-2030 KPI	ITU-R Target / Objective	AI/ML-Driven Enablers in 6G	Expected Improvement / Outcome	Representative References
Peak Data Rate & Throughput	Up to 1 Tbps (> 20× 5G)	DL-based beamforming, intelligent RIS control, generative channel modeling, adaptive modulation	> 50 % spectral-efficiency gain; adaptive link capacity under dynamic conditions	[32–59]
Latency & Reliability (URLLC / HRLLC)	≤ 0.1 ms E2E latency / 99.99999 % reliability	RL-based scheduling & network slicing, predictive handover, edge-AI inference placement	Sub-ms decision loops; autonomous error-recovery and QoS assurance	[60–62]
Energy Efficiency / Sustainability	> 10× energy efficiency vs 5G baseline	AI-based load forecasting, sleep mode control, edge collaboration optimization	30–40 % network-level power reduction; extended battery life for IoT nodes	[63,64]
Spectral Efficiency & Capacity Density	> 15 bit/s/Hz per cell	Cognitive radio with RL control, GNN-based topology learning, semantic compression	Dynamic reuse and semantic transmission increase effective capacity by > 2×	[65,66]
Massive Device Connectivity (mMTC)	> 10^7 devices /km^2	Federated learning for distributed access, MARL for resource sharing, auto-scalable schedulers	Near-linear scaling without centralized signaling overhead	[67,68]
Positioning & Sensing Accuracy	< 1 cm localization / 3D sensing	AI-based ISAC waveform design, multi-modal sensor fusion, generative signal estimation	25–50 % accuracy gain under multipath and NLOS scenarios	[51–69]
Security & Trustworthiness	Resilient & explainable network operation	XAI models, blockchain-audited FL, adversarial training for robustness	Attack detection > 95 %; verifiable decision traceability	[70,71]
Intelligence Level (AI-Native Metric)	Self-learning / self-optimizing network behavior	Digital twins, semantic reasoning agents, continual learning at edge	Autonomous adaptation to context → "zero-touch" operation	[72,73]

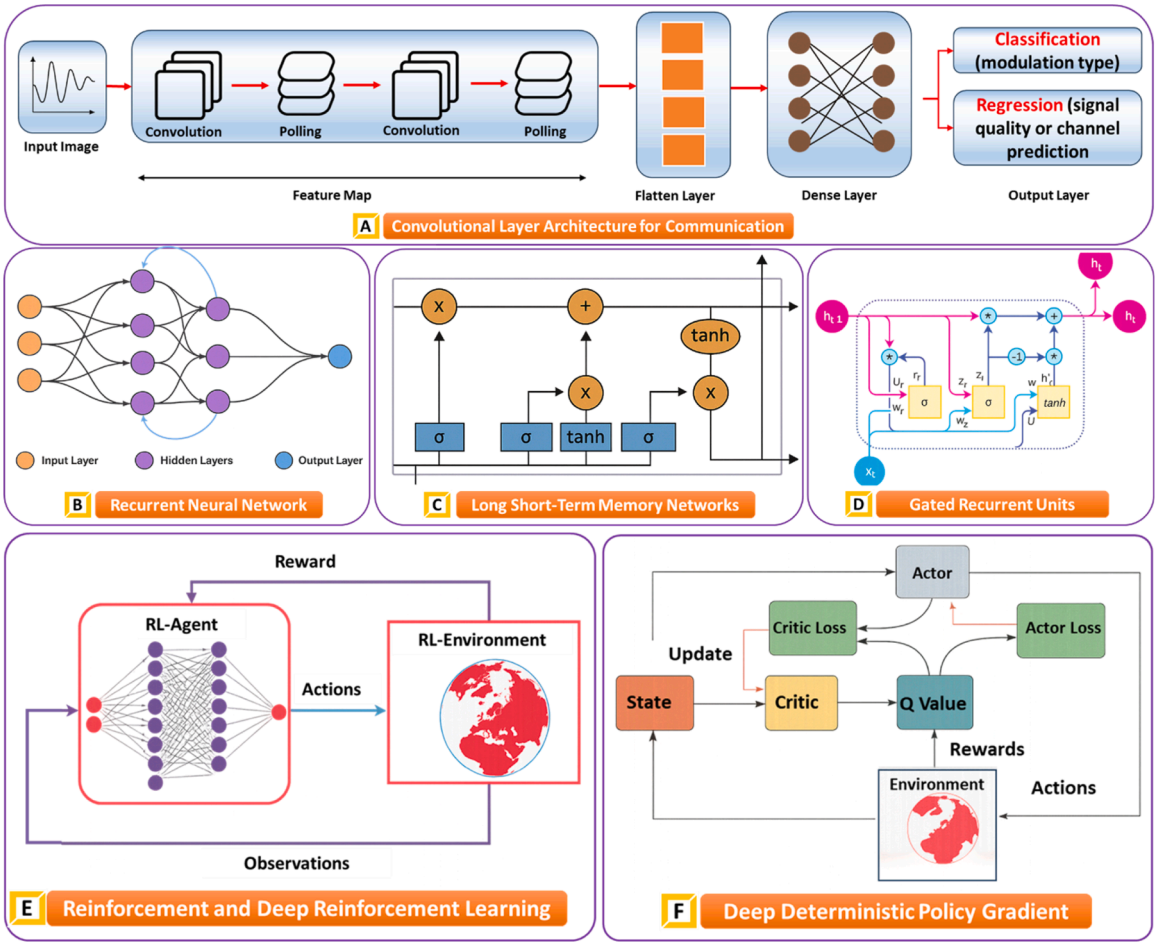


Fig. 2. (A) A CNN-based architecture for wireless communication applications. (B) A basic RNN structure where hidden layers retain temporal dependencies from input sequences. (C) An advanced RNN variant that uses memory cells and gating mechanisms to handle long-term dependencies in sequential data, making it suitable for dynamic signal and channel modelling. (D) A simplified version of LSTM that combines the forget and input gates into a single update gate, reducing computational complexity while preserving the ability to model temporal sequences effectively. (E) A generic framework showing interaction between an RL agent and its environment. (F) An advanced actor-critic algorithm where the Actor proposes actions, and the Critic evaluates them using Q-values.

temporal memory allows for robust predictions even in the presence of delayed or missing data, making them suitable for delay-sensitive and mobility-driven scenarios in 6G, as illustrated in Fig. 2(C) [81,82].

3.1.4. Gated recurrent units (GRUs)

GRUs simplify LSTMs and therefore have fewer parameters, yet yield comparable results. This is beneficial for applications in which

computational cost is key, e.g. in real-time processing, for example in mobile devices or at the network edge. GRUs have been applied to HARQ (Hybrid Automatic Repeat Request) prediction, packet arrival modelling, and energy demand forecasting in wireless networks, providing a good trade-off between performance and complexity, as illustrated in Fig. 2(D) [83,84].

3.2. Reinforcement and deep reinforcement learning

Reinforcement Learning (RL) offers an effective, biologically inspired tool for sequential decision-making in dynamic and partially observable wireless environments, environments that are typical of 5G and 6G systems. The agent interacts with the environment by performing actions, monitoring the resulting states, and receiving feedback in the form of rewards or penalties (Fig. 2 E). Through repeated interactions, the agent learns the optimal policy, which is the policy that maximises the expected cumulative reward with the passage of time. The trial-and-error learning scheme is particularly suitable for complex radio networks in which explicit mathematical models either do not exist or are computationally intractable. RL has therefore been widely adopted for tasks such as dynamic spectrum access, power and rate control, user association, mobility management, and network slicing [85–87].

3.2.1. From classical RL to deep reinforcement learning

Methods for traditional RL such as Q-Learning and SARSA which employ discrete state-action tables for estimating Q-values (i.e to identify the utility of performing an action in a given state). Though simple and interpretable, these tabular approaches do not scale well when the state space becomes large or continuous, such as in the application of massive MIMO beamforming or heterogeneous network coordination [88]. Deep Reinforcement Learning (DRL) responds to this constraint by integrating RL principles with deep neural networks to represent value and policy functions. In doing so, it extends the adaptability of reinforcement learning to high-dimensional and non-linear problems of next generation wireless systems iii [89]. Fig. 2 (F) illustrates a typical actor–critic architecture where the actor proposes actions and the critic evaluates them based on predicted returns, iteratively refining the policy for continuous control tasks. The Deep Deterministic Policy Gradient (DDPG) and Proximal Policy Optimization (PPO) algorithms are two prominent DRL methods that have demonstrated impressive performance in wireless applications. DDPG can easily deal with the continuous action spaces such as adjustment of transmit-power or steering of antennae. On the other hand, PPO is able to perform stable learning in large multi-agent settings through the use of trust-region based policy updates [90–92].

3.2.2. Multi-Agent deep reinforcement learning for decentralized 6G resource allocation

As 6G networks evolve toward massive decentralization and heterogeneous topologies, single-agent DRL architectures struggle to capture the spatial and temporal interdependencies among network entities. Multi-Agent Deep Reinforcement Learning (MARL) extends the DRL framework by deploying multiple cooperative or competitive agents that interact with each other and with their local environments. Each agent represents a decision-making entity, such as a base station, small cell, UAV, or IoT device, and learns local policies based on partial observations while a global objective emerges from collective behavior. Through parameter sharing and limited message exchange, MARL achieves distributed learning and coordination without the overhead of a centralized controller [40–93].

In decentralized 6G scenarios like device-to-device (D2D) communication, vehicular V2X networks, and cell-free massive MIMO-MARL, there are several inherent advantages over classical optimization techniques. The service can learn autonomously, i.e., interference-aware power control, channel selection, or task-offloading policies can be applied under non-stationary conditions, by assigning an agent to each node or cluster [91,94]. Actor–critic variants like MADDPG (Multi-Agent Deep Deterministic Policy Gradient) and multi-agent PPO provide robust convergence even in partially observable environments, balancing local autonomy and global coordination. The MARL agents can enable real-time, low-latency resource allocation and spectrum reuse with negligible signaling overhead in RAN Intelligent Controllers (RICs) or edge orchestrators. This will indeed be an important milestone toward achieving self-organizing, scalable, and energy-efficient operation as envisioned for AI-native 6G networks [95].

3.3. Graph neural networks (GNNs)

GNNs have arisen as a new type of deep learning paradigm able to model complex, non-Euclidean data structures, precisely the type of topologies appearing in wireless communication networks. GNNs can learn over arbitrary structured graphs where the nodes of the graph represent entities like base stations, users and devices onto which links represent other aspects like connectivity, interference or mobility patterns. This graph modelling aspect is what enables GNNs to be particularly suited to 6G wireless systems which are mostly decentralised, dynamic and heterogeneously connected. GNNs enable the network to reason over both local and global dependencies, making them ideal for tasks involving spatial coordination, cooperative behaviour, and multi-agent learning [96,97].

3.4. Generative models for synthetic channel modelling

Generative models like Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) are proven effective at learning the true distributions of the underlying data and synthesizing realistic artificial data. In wireless communication, such models are important for simulating complex high-dimensional environments in which collecting large quantities of high-quality real data is difficult/impracticable [98,99]. Applications range from synthetic channel modelling and CSI generation to adversarial testing, anomaly injection, and enhancing training datasets for supervised learning tasks. As 6G envisions AI-native architectures relying heavily on data-driven intelligence, generative models help alleviate the data scarcity problem, especially for rare conditions like edge fading scenarios or low-SNR environments [100,101].

3.4.1. Generative adversarial networks (GANs)

The fundamental design of GANs consists of two collaborating neural networks: a generator which produces data and a discriminator which grades the produced data as authentic. Through adversarial training, the GANs learn how to produce realistic samples indistinguishable from natural data. This is illustrated in Fig. 3(A). Wireless communication is one of the more prominent applications. Instead of relying upon handcrafted statistical models such as Rayleigh or Rician fading, GANs will be able to learn the joint distribution of complex wireless channels through empirical measurement or simulation of the wireless channels, thereby permitting the construction of data-driven channel emulators [102,103]. These emulators can be fine-tuned for specific environments (e.g., urban, indoor, vehicular) and are valuable for testing AI models in realistic yet synthetic settings. GANs have also been utilized for modulation synthesis, anomaly injection, and data augmentation in deep learning-based receiver training. Additionally, conditional GANs facilitate enhanced control by conditioning the generation process on certain labels or inputs, such as user mobility, frequency bands, or antenna configuration, thus rendering them effective for custom synthetic dataset generation in 6G simulation environments [104,105].

3.4.2. Variational autoencoders (VAEs)

VAEs offer a probabilistic approach to generative modelling by learning to encode input data into a latent distribution and then decode

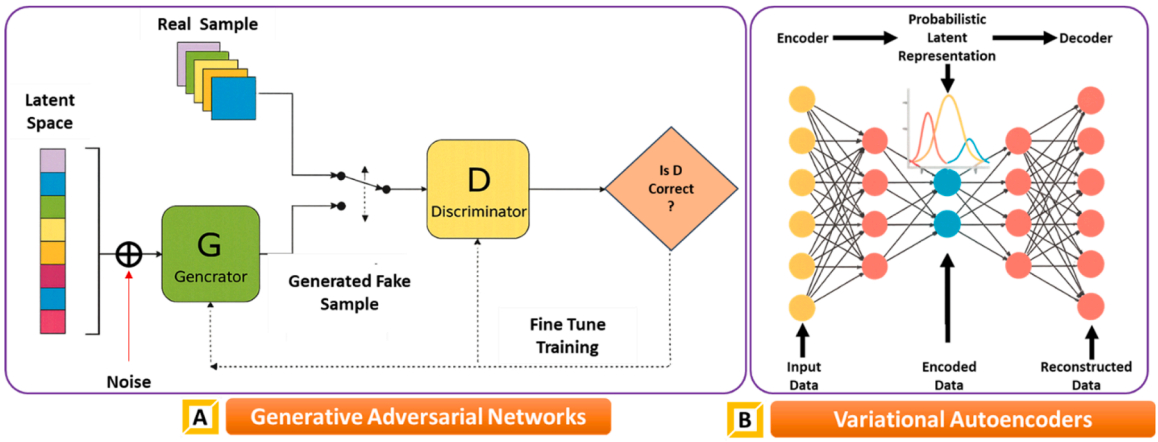


Fig. 3. (A) GANs consist of a Generator and Discriminator trained adversarially to create realistic synthetic data from latent noise. (B) VAEs Encode input into a probabilistic latent space and decode it for data reconstruction, enabling smooth interpolation and denoising.

it back into a realistic reconstruction. Unlike GANs, VAEs provide an explicit likelihood model and are more stable during training. In wireless communication, VAEs have been utilized for channel state estimation, pilot contamination mitigation, and compressive CSI feedback by learning compact representations of high-dimensional data, as shown in Fig. 3 (B) [106,107]. VAEs are particularly advantageous when interpretability and structure in the latent space are important, such as for semantic communication, where messages are transmitted in compressed latent representations, or radio map generation, where VAE-based interpolation fills coverage gaps in spatial signal strength data. In 6G systems, where joint source–channel encoding and task-oriented communication are emerging paradigms, VAEs offer promising capabilities for learning semantic-level representations aligned with downstream inference tasks [108–110].

3.4.3. *Complementary roles and hybrid approaches*

While GANs excel at generating high-fidelity samples, they lack explicit density modelling. Conversely, VAEs offer better interpretability and probabilistic reasoning but may produce blurrier outputs. To combine their strengths, hybrid models such as VAE-GANs have been proposed and applied to wireless domains requiring both generative quality and inference capability. Additionally, generative models are increasingly being used in unsupervised pretraining, adversarial robustness testing, and zero-shot learning in scenarios where labelled data is scarce or task-specific datasets are unavailable. In 6G's digital twin environments, generative models can simulate entire network states, device trajectories, or RF landscapes to support AI agents in training, testing, and policy exploration without interfering with real-world networks [95].

Different AI and ML models have different compatibility with architectures depending on computational intensity, data privacy, and degree of coordination needed. Deep and generative models, i.e. CNNs and GANs and VAEs, are best trained and inferred in centralized infrastructures in the cloud where computing and memory resources are plenty. In contrast, sequence learning and control-models, i.e. RNNs, LSTMs, and RL models, fit best in the distributed or hybrid environments

Table 3
AI/ML model suitability, applications, and control functions across 5G/6G architectures [113–117].

Model / Technique	Primary Strengths	Typical Wireless / Network Application	Controlled Variable / Operational Impact	Suitable Architecture	Remarks / Key Insights
CNN / DNN	Spatial feature extraction; high accuracy	Beamforming, modulation recognition	Updates beam-selection codebook index or CSI map for optimal beam alignment	Centralized (cloud training + inference)	High compute demand; ideal for base-station servers and mmWave beam selection
RNN / LSTM / GRU	Temporal sequence learning	Mobility prediction, traffic forecasting	Adjusts handover thresholds and mobility timers based on sequential inputs	Distributed / Hybrid (edge-enabled)	Suitable for latency-critical forecasting; deployed at edge or hybrid nodes
Reinforcement Learning (RL / DRL)	Adaptive decision-making under uncertainty	Dynamic spectrum access, power control	Learns optimal policies to maximize long-term reward under stochastic conditions	Distributed / Hybrid (edge agents)	Enables self-organizing and real-time resource allocation with minimal signaling
DDPG (Deep RL)	Continuous control in high-dimensional state spaces	Reconfigurable Intelligent Surface (RIS) tuning	Modifies reflection phase vector Φ of RIS elements to maximize SNR	Distributed Edge	Supports RIS-assisted channel enhancement for adaptive environment control
PPO (Policy Gradient RL)	Stable policy optimization with large action spaces	Network slicing / resource allocation	Adjusts slice-admission rules and bandwidth quotas based on QoS rewards	Edge / Hybrid	Efficient for continuous adaptation of slice QoS and bandwidth fairness
Graph Neural Networks (GNNs)	Models non-Euclidean topologies and dependencies	Inter-cell coordination, D2D/V2X scheduling	Learns connectivity graph; activates optimal link sets in real-time	Distributed (edge controllers)	Learns spatial correlations for scalable, graph-aware wireless coordination
Generative Models (GAN / VAE)	Synthetic data generation, data augmentation	Channel emulation, anomaly injection	Generates synthetic CSI or rare-channel conditions for model training	Centralized / Cloud (offline)	Computationally intensive but useful for data scarcity and channel modeling
Federated Learning (FL)	Privacy-preserving collaborative training	Vehicular and Healthcare IoT	Updates global model weights θ = Σ wiθi without sharing raw data	Federated Edge + Cloud	Enables privacy-compliant distributed learning; efficient in large IoT networks
Split Learning (SL)	Layer-wise training split between client and server	Edge–cloud collaborative AI	Divides model across devices and servers to balance load and security	Hybrid (edge + cloud)	Reduces edge device burden; preserves privacy through intermediate representation sharing

now typical of the edge, where real-time decision-making and latency are of concern. Federated learning and split learning models are naturally suited for federated infrastructures where privacy and decentralisation and inter-device learning are important. This mapping ensures the practical alignment of algorithmic design with architectural deployment considerations in AI-driven 5G/6G systems [111–113]. To clarify the applicability of training the models to deploy in operations, the data in Table 3 now clearly defines for each of the AI/ML models the best suited architecture to consider as a distribution method, that being whether centralized, distributed or federated, depending on computational complexity, data sharing requirements and latency. From Table 3 it can be seen how deep models and generative models best suit deployment in centralized cloud environments with their high compute requirements, while sequence learning models and control oriented models are best suited to distributed or hybrid edge environments that require real time adaptations of their neural networks. Privacy-sensitive learning paradigms like Federated and Split Learning are inherently suited for federated architectures, enabling decentralized intelligence while preserving data confidentiality.

4. Key evaluation metrics in AI/ML-Enabled 5G/6G systems

As AI/ML are integrated throughout the wireless communication stack, including physical layer signal processing and service-level orchestration, they must also be evaluated based on a multi-dimensional set of metrics [118,119]. In contrast to traditional networks that depend only on deterministic or data throughput indicators, AI/ML wireless systems require model, network, and system level evaluation to holistically capture effectiveness, reliability, efficiency, and deployability. This section categorizes and explains the most relevant metrics used to evaluate AI-driven wireless solutions.

4.1. Model-Level metrics

These measures evaluate the algorithmic performance of AI/ML models independently of the network. These are typically employed in the model training, validation, and testing phases. In fact, the performance of AI/ML models in wireless communication should first be evaluated in terms of such model-level measures as accuracy, precision, F1-score, mean square error (MSE), inference time and model size. These metrics provide insight into the algorithm's standalone performance before integration into the broader network (see Table 4).

Table 4
Model-level evaluation metrics in AI/ML-enabled wireless systems [120–124].

Metric	Purpose	Application in Wireless
Accuracy, Precision, Recall, F1-Score	Measure classification performance and prediction reliability.	Signal classification, modulation recognition, spectrum sensing.
AUC (Area Under ROC Curve)	Evaluate classifier's ability to distinguish between classes.	Anomaly detection in network traffic, intrusion detection.
MSE / RMSE	Measure error in regression tasks or reconstruction fidelity.	Channel estimation, CSI compression, trajectory prediction.
Training Time & Inference Time	Evaluate time-efficiency of model development and deployment.	Critical for real-time communication applications.
Model Size (Parameters, Memory Footprint)	Reflects resource consumption, especially for edge deployment.	Important in FL, SL, or edge-AI for lightweight inference.
Energy Consumption	Total power required during model training and inference.	Essential for battery-powered devices, UAVs, and IoT nodes.

4.2. Network-Level metrics

When AI/ML models are deployed within wireless systems, their effects on network-level performance evaluations should be based on parameters such as traffic delay, throughput, energy efficiency, and handover stability. These parameters are essential to provide assurance that the integration of AI will enhance the levels of end-to-end communication quality and reliability (see Table 5).

4.3. System-Level metrics

In addition to performance, AI-driven wireless systems must be evaluated with regards to their system performance as it relates to aspects such as scalability, explainability, privacy, and cost of deployment. These measurement dimensions are important for demonstrating effective real-world efficacy, trustworthiness, and success in operational utility over the long term (See Table 6).

4.4. AI process efficiency metrics

While model- and network-level measures gauge performance and quality, equally important is the efficiency of the AI/ML pipeline itself, particularly in constrained environments such as on-device and edge deployments. The proposed metrics giving an indication of computational sustainability in terms of AI/ML processes in 5G/6G networks are summarized in Table 7.

5. Key use cases and applications

5.1. Autonomous vehicles and vehicle-to-everything V2X communication

In order to improve vehicle safety, coordination, and decision-making in dynamic environments, autonomous vehicles (AVs) use V2X communication (vehicle-to-everything) as well as the sensor capabilities that they have on board. V2X encompasses not only vehicle-to-vehicle (V2V) interaction, but also vehicle-to-infrastructure (V2I), vehicle-to-network (V2N) and vehicle-to-pedestrian (V2P) interactions to form an interconnected ecosystem for mobility and cooperative driving for autonomous veichles. The onboard sensors provide situational awareness, but are limited in range and by their surroundings; V2X enhances

Table 5
Network-level evaluation metrics in AI/ML-enabled wireless systems [77,125].

Metric	Purpose	Application in AI-Enabled Networks
Latency & Jitter (E2E Latency)	Measures total end-to-end delay (UE → core → application server), including air-interface, backhaul/transport, and processing latency, plus variation in delay.	Critical for ultra-low-latency use cases such as telesurgery, extended reality (XR), and V2X communication; minimized using AI-driven predictive scheduling, dynamic offloading, and split inference.
Throughput	Measures data transfer rate across the network.	Optimized by AI-driven resource scheduling, dynamic routing.
Packet Loss Rate	Indicates reliability and congestion handling.	AI-based congestion control and handover decision-making.
Spectral Efficiency	Bits transmitted per Hz of bandwidth.	Enhanced via AI-optimized beamforming, RIS configuration.
Energy Efficiency	Data transmitted per unit energy consumed.	AI-managed power allocation and sleep scheduling in dense deployments.
Handover Success Rate	Stability of user transitions across cells.	Predictive AI models minimize dropped calls in mobile scenarios.
Fairness Index (e.g., Jain's Index)	Resource fairness across users or services.	Important in multi-user scheduling and QoS-aware ML models.

Table 6
System-level metrics in AI/ML-enabled wireless systems [126,127].

Metric	Purpose	Application Context
Quality of Experience (QoE)	Subjective user satisfaction with service quality.	AI personalization, adaptive bitrate control, edge inference.
Scalability	System's ability to handle more devices/users without degradation.	Federated Learning across millions of edge devices.
Explainability	Clarity and interpretability of model decisions.	Required for safety-critical systems like autonomous vehicles or telesurgery.
Privacy Score	Quantitative or qualitative privacy assurance.	FL, SL, and differential privacy in sensitive applications.
Deployment Cost	Includes infrastructure, training, and operational expenses.	Choosing between centralized, edge, or hybrid AI deployments.
Fault Tolerance	System resilience to failures or unexpected behavior.	AI-enabled self-healing networks and redundancy planning.

this feature by allowing vehicles to share such information as intent, trajectory and perceived environmental information in real time [132–134]. AI/ML techniques are fundamental to decoding the high velocity, high volume V2X data for prediction, optimization and decision making. Deep learning models (e.g., CNNs for vision tasks), reinforcement learning (for dynamic routing) and predictive analytics (e.g., QoS estimation) would find perennial applications over protocol stacks and at the edge-cloud. As 6G envisions even denser environments with high mobility, intelligent edge computing and proactive resource orchestration enabled by AI would be crucial to address the demands of ultra-low latency and reliability in AV applications [135,136].

Recent studies provide strong support for how AI/ML techniques integrated into V2X communication systems improve performance in autonomous vehicles in safety, perception, resource optimization, and real-time decision making. Ma et al., in a prototype MEC-aided 5G-V2X system, developed a prototype system to overcome the challenges of high latency and traffic congestion in cooperative driving scenarios.

They proposed a deep spatiotemporal residual network (PST-ResNet), specifically developed for forecasting city-scale traffic, in combination with a SC-DSCS algorithm based on swarm intelligence for dynamic optimization as illustrated in Fig. 4 (A). These AI models reduced operational costs as well as improving platooning stability over conventional approaches based on the use of sensors alone [137]. This study effectively demonstrates the role of AI in enabling scalable, latency-aware vehicular coordination within MEC-integrated 5G/6G frameworks. Targeting the secure exchange of V2X data, Weerasinghe et al. tackled the significant problem of transient trust creation in 5G-based V2X networks. Their solution, a key exchange protocol based on threshold cryptography, allows for rapid, secure and anonymous communication even in brief vehicle interactions (see Fig. 4(B)). Unlike traditional static certificate-based systems, this is certainly more suited to dynamic topologies of vehicles with different ownership characteristics [138]. The work highlights the importance of cryptographic trust frameworks for privacy-preserving and low-latency vehicular communication in evolving 6G architectures. In a contrasting vein, Xu et al. created a lane detection system driven by 5G-V2X along with a Mask R-CNN improvement, rendering the road feature identification system very powerful and real-time. (See, for instance, Fig. 4(C)). Their method revealed the limitations of classic methods employing Hough transforms in occluded and dimly lit urban streets by increasing detection accuracy and processing speeds [139]. This study demonstrated how deep visual models, when integrated into the V2X stack, enhance perception accuracy and resilience under challenging driving conditions.

As far as predictive safety is concerned, Al-Yarimi's SafeRouteX framework is based on LSTM-VAE to extract features, Elastic Net for dimensionality reduction, and Random Forest for accident predictions in V2X environments. This ensemble was found to outperform simple statistical models with regard to accuracy and generalizability [142]. The presented study reinforces the role of hybrid AI architectures in proactive accident avoidance within intelligent transportation systems. Similarly, Ribeiro et al. proposed a stacked LSTM architecture to predict collisions involving Vulnerable Road Users (VRUs), particularly motorcyclists, as shown in Fig. 4 (D). While the model achieved high prediction rates, its relatively high false-positive count highlighted the difficulty of applying automated braking in safety-critical systems [140]. On the signal processing front, Liu et al. introduced a sliding-window Fourier transform-based ensemble ML model for real-time V2X signal analysis, demonstrating superior energy efficiency and communication reliability compared to conventional modulation-based systems [143]. The demonstrates the potential of data-driven spectral analysis to enhance energy-efficient V2X communication for next-generation vehicular networks. Quality of service (QoS) demands a proactive management approach. Barmpounakis et al. devised a lightweight artificial intelligence (AI)-driven tool, PreQoS, for predicting end-to-end, QoS parameters like delay and data rate in V2X applications (see Fig. 4 (E)). The predictive nature of this application allows for pre-planned modulation of resources available to the network and thus facilitates improved continuity of applications as opposed to reactive network systems [141]. The PreQoS study by Barmpounakis and colleagues illustrates how predictive AI models can dynamically optimize QoS provisioning to ensure seamless vehicular connectivity in 6G-enabled V2X systems.

The provision of stringent QoS for thousands of concurrent vehicular links remains a major challenge in dense V2X and D2D scenarios. Although the prediction of the network condition is valuable, safety-critical services like cooperative collision avoidance need latency guarantees (<10 ms) and ultra-high reliability (> 99.999 %). Recent developments go further to embed these constraints directly into AI/ML learning objectives [144]. This includes delay penalties and reliability rewards in constrained-aware RL, so agents optimize for throughput and continuity of service. Graph neural networks and federated multi-agent RL frameworks capture the underlying spatiotemporal correlations

Table 7
AI process efficiency metrics for edge and on-device intelligence [128–131].

Metric	Definition / Focus	Importance for 5G/6G AI Systems	Typical Measurement Unit / Example
Training Time	Duration required for model convergence during learning.	Shorter training enables rapid model adaptation to new channel or traffic patterns.	Seconds / minutes per epoch; total wall-clock time (e.g., 120 s/epoch)
Inference Latency	Delay between input data reception and model output.	Determines real-time response capability in V2X, XR, and telesurgery.	Milliseconds per inference (e.g., < 5 ms)
Energy Consumption	Power required during model training and inference.	Critical for energy-limited IoT and UAV devices; affects battery life and thermal limits.	Joules per inference / Watts per epoch
Model Size / Memory Footprint	Storage required for weights and intermediate activations.	Impacts deployability on small edge processors and microcontrollers.	Megabytes (MB) or number of parameters
Compute Load / FLOPs	Number of floating-point operations per inference.	Quantifies computational efficiency; lower FLOPs = lower latency and cost.	GigaFLOPs or operations/inference
Thermal Efficiency	Ratio of useful computations to heat dissipation under sustained workloads.	Ensures model reliability on embedded or fanless devices.	°C rise per watt; efficiency (%)

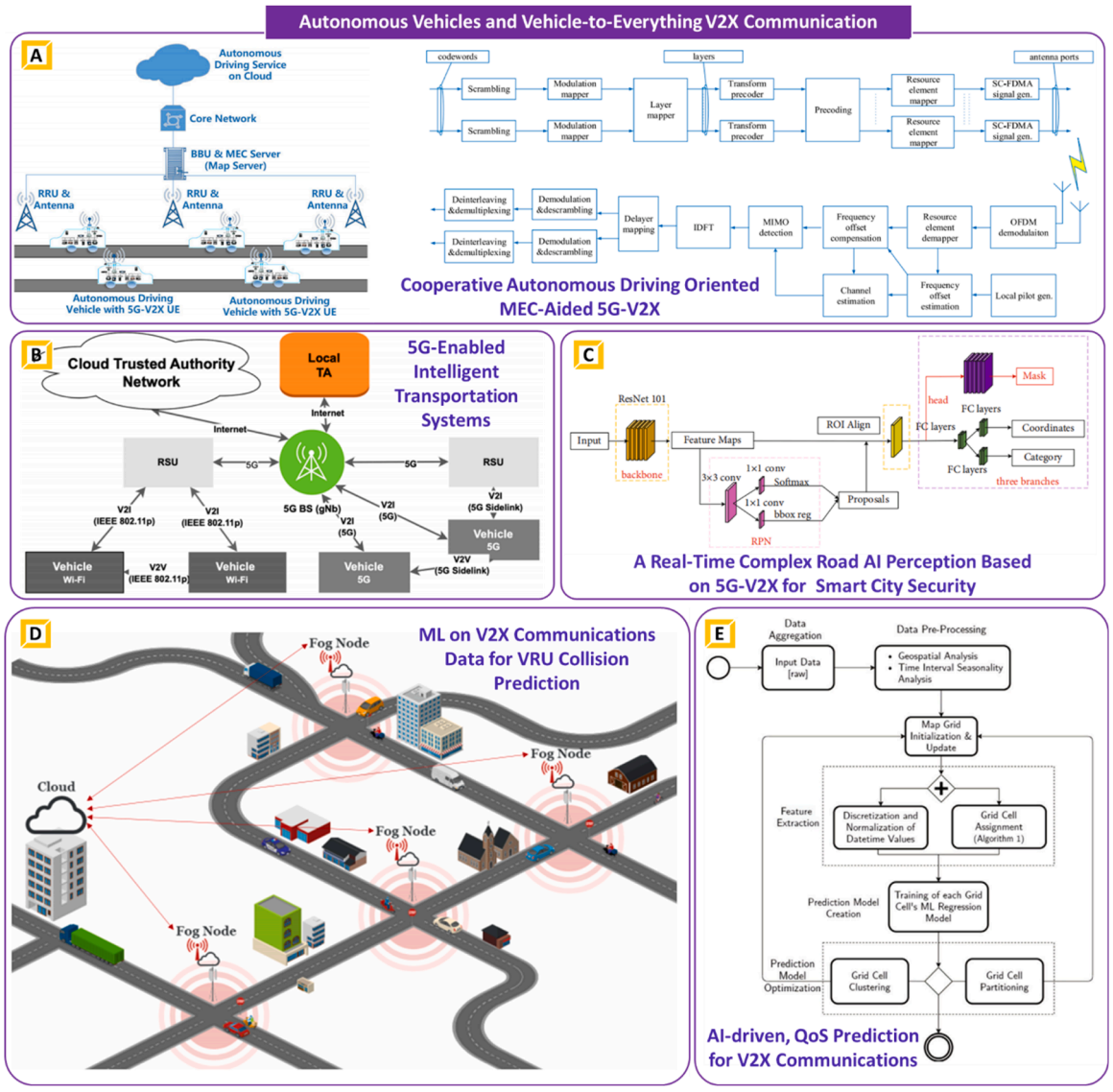


Fig. 4. (A) Cooperative autonomous driving supported by Mobile Edge Computing (MEC) and 5G, enabling low-latency communication between vehicles and infrastructure, taken from [137] with the permission of IEEE. (B) A secure cloud–fog–vehicle framework using RSUs and 5G base stations for seamless V2X and vehicle-to-cloud communication, taken from [138] with the permission of MDPI. (C) Deep learning pipeline (e.g., ResNet101) for object detection and segmentation to enhance traffic awareness using 5G-V2X, taken from [139] with the permission of Wiley. (D) ML on V2X data and fog nodes to predict collisions involving vulnerable road users in smart urban environments, taken from [140] with the permission of MDPI. (E) Grid-based prediction model using spatial-temporal data to optimize quality of service (QoS) in vehicular communication, taken from [141], with the permission of Elsevier.

among the moving vehicles and adaptively update the scheduling and power control strategies for maintaining link reliability. By integrating these QoS-aware agents into RAN Intelligent Controllers (RICs) or edge orchestrators, future 6G networks can proactively allocate spectrum, computing, and backhaul resources to ensure predictable performance even in ultra-dense traffic [145,146]. Complementary advancements have clearly shown how AI/ML is transitioning V2X asset management from reactive utilization to proactive and potentially predicted management. However, practical deployment demands the right amount of optimization accuracy against computational load. While deep RL and federated models provide near-optimal scheduling, they have higher processing and synchronization requirements and contribute additional signalling load - all of which, especially in real-time multi-agent applied contexts, can be higher still. Moreover, modern designs tend to opt for lightweight and compressed models and cognitively offload concepts such as split-inference and edge collaborative training schemes that only add less than a millisecond latency overhead with lower power and bandwidth required and higher scalability. Accordingly, to that end, a holistic architecture that co-optimizes algorithmic complexity, QoS guarantees, and network resources is recommended. Future research should pursue complexity-aware AI frameworks capable of adaptively scaling with mobility, density, and reliability requirements, ensuring both performance efficiency and computational viability across next-generation 6G V2X deployments [147–149].

5.2. AI/ML frameworks for dense device-to-device (D2D) communication

Dense in-band D2D underlay amplifies co-channel interference because multiple D2D pairs often reuse the same cellular user (CU) resource block, creating tightly coupled interference graphs where centralized radio-resource management (RRM) incurs heavy signalling and slow convergence. Distributed AI offers a scalable alternative. Ioannou et al. propose a componentized Distributed-AI (DAI) framework (see Fig. 5A) in which local learners coordinate via lightweight

messages, adapting power/channel choices to topology changes with markedly lower control overhead than centralized baselines [25]. In parallel, pricing/Stackelberg schemes transform interference into an economic signal, the BS (leader) announces interference prices for cellular user (CU) resource reuse, while D2D pairs (followers) choose power/sub-channels to maximize their utilities, yielding higher throughput with bounded CU harm in two-tier deployments (see Fig. 5B), recent variants use non-uniform pricing and joint sub-channel/power designs to improve fairness and stability [150,151]. Complementing these, distributed DRL/MARL learns reuse and power policies under partial observability (see Fig. 5C), DRL underlay schemes have demonstrated significant gains in spectral efficiency and QoS guarantees, while multi-agent PPO methods further improve sum-rate and slice-aware fairness without a central scheduler [152].

Traditional gradient-based adaptive algorithms depend on explicit, differentiable cost functions and accurate channel models. Such assumptions often fail in 6G environments characterized by stochasticity, non-convex objectives, and partial observability. By contrast, DRL learns model-free policies directly from interaction and reward signals, removing the need for analytic gradients. This capability allows DRL to adapt to unknown or time-varying channel dynamics, making it more robust under real-world wireless conditions. Moreover, DRL naturally supports multi-agent coordination and decentralized learning, crucial for 6G edge deployments where agents must make context-aware decisions in real time. Hence, DRL is promoted as a superior approach for self-organizing, latency-sensitive, and energy-efficient D2D resource management.

Finally, graph-neural schedulers operate directly on interference graphs to select simultaneous D2D links by learning graph embeddings that capture pairwise conflicts. GNN-based policies reduce collisions and boost link success in highly mobile or multi-hop D2D settings compared with heuristic graph coloring, as shown in Fig. 5D [153]. Collectively, these distributed and pricing-based approaches scale with the number of active pairs, curtail backhaul/control signalling, and maintain CU protection, addressing the very limitations that hamper centralized controllers in ultra-dense D2D reuse [154,155].

5.3. AI-Enabled smart healthcare and telesurgery

The convergence of AI and next-generation wireless communication technologies is transforming the healthcare landscape, particularly in the domains of smart diagnostics and telesurgery. As 6G networks promise ultra-low latency, ubiquitous connectivity, and massive device integration, they enable advanced medical applications that demand real-time responsiveness, precision, and contextual awareness [156,157]. AI models, ranging from deep learning to federated optimization, are increasingly integrated into cyber-physical healthcare systems to process massive volumes of multimodal data from wearable sensors, imaging devices, and EHRs. Besides integrated thermal imaging and sensor data, imaging can also be achieved through wireless signal reflections using channel-state-information (CSI)-based sensing [158]. This paradigm shift permits monitoring and personalized interventions, enabling predictive analytics to prevent extreme nested occurrence, such as a cardiac arrest, and automate telesurgery and remote monitoring.

In contrast to earlier systems that were rigid and rule-bound in communication architecture, AI-6G in health architectures deploys edge intelligence, digital twins and cybercare infrastructures for decentralized, dynamic, and resilient healthcare decision support across the physical and digital delivery of critical care. Recently, there is an emerging literature reporting a variety of AI use case studies on the transformational impact of 6G technologies across the smart healthcare continuum. For example, Hadi et al. applied an ensemble learning framework (Naïve Bayes, Decision Tree and, Logistic Regression) framework applied in a cognitive HetNet for real-time stroke prediction and PRB allocation as illustrated in Fig. 6 (A). This MILP-based optimization ensured that the outpatients expressing stroke symptoms got prioritized access to the network, finally reducing the critical delays by up to 95 %. The presented study effectively illustrated how AI-enhanced scheduling within 6G HetNets can drastically improve patient triage and emergency responsiveness [159].

Moving to resource management, Syed et al. suggested a Fuzzy Neuro-based Cybertwin system to coordinate healthcare in the edge-cloud environment, depicted in Fig. 6 (B), so this distributed learning model improved resource utilization and minimized transmission costs to the cloud significantly better than static load allocation techniques. Their study highlights how AI-driven Cybertwin frameworks can dynamically orchestrate computational tasks between edge and cloud for real-time medical analytics [160]. Furthermore, Khan et al. developed a big data analysis framework, which in addition to 6G, also includes AI. This real-time diagnostic and monitoring framework is illustrated in Fig. 6 (C). The model validated on various throughput, latency and error metrics, is safe and effective, yielding excellent results in fast delivery and good decision-making in resource-poor environments. Khan's work exemplifies the feasibility of integrating 6G's ultra-reliable low-latency communication with AI pipelines for timely diagnostic insights [161]. In addition, Elaziz et al. implemented an image classification system built using MobileNetV3 with a novel optimization technique, AOAHG, achieving over 99 % accuracy with OCT and WBC datasets in a 6G-IoMT setup as seen in Fig. 6 (D). This work demonstrates how lightweight deep models can leverage 6G bandwidth to enable cloud-assisted diagnostics with minimal delay and exceptional precision [162].

In telesurgery applications, Shah et al [163]. proposed a secure, AI-orchestrated garlic routing system using GRU models and blockchain-encrypted routing tunnels, achieving 96.89 % accuracy in identifying malicious transmissions, ensuring safe robotic operations, shown in Fig. 6 (E). Their framework underscores the role of AI-enabled trust mechanisms in securing latency-critical telesurgery networks. Thanganadar et al [164]. designed a cyber-physical healthcare model integrating quantum optimization and federated honeypots for intrusion mitigation, achieving a MAP of 97 % and network security of 96 %. This study reflects how quantum-inspired federated AI can fortify next-generation healthcare systems against cyber-physical attacks. Similarly, Manikandan et al. used quantum-dot-enabled optoelectronic sensors and ResNet models for cardiac disease detection, reaching 98 % accuracy [165]. Lastly, Alshammari et al. presented a decentralized Feed-FDNN with ABC optimization for patient data prediction, achieving improved inference time and diagnostic precision in patients. These various frameworks are indicative of a great breadth and depth of AI-6G synergy in contemporary medicine, ranging from prediction and diagnosis to secure surgery [166]. This contribution demonstrates how bio-inspired optimization and federated AI can enhance reliability and responsiveness in distributed healthcare networks.

While these advances represent great strides, many challenges remain. Many AI models are subject to overfitting, data privacy issues, and an over-reliance on simulation or synthetic datasets making them less generalizable against real clinical noise. The telesurgery systems are hindered by issues of cyber-physical security, test of the integrity of the devices, and regulatory inconsistencies at different locations. Real-time diagnostics require compression of model structures and low-power inference structures but many deep architectures remain computationally strong. Furthermore, ethical issues such as explainability, liability from the autonomous procedures, and patient consent remain unresolved. Future studies should explore hybridised AI systems that include symbolic reasoning and deep learning, federated and transfer learning tides towards greater individualisation in the presence of privacy issues, and quantum-aggrieved AI for faster convergence of models. The integration of digital twins, cybercare avatars, and edge-based explainable AI will likely shape the next frontier of intelligent, equitable, and secure healthcare in 6G environments [167–169].

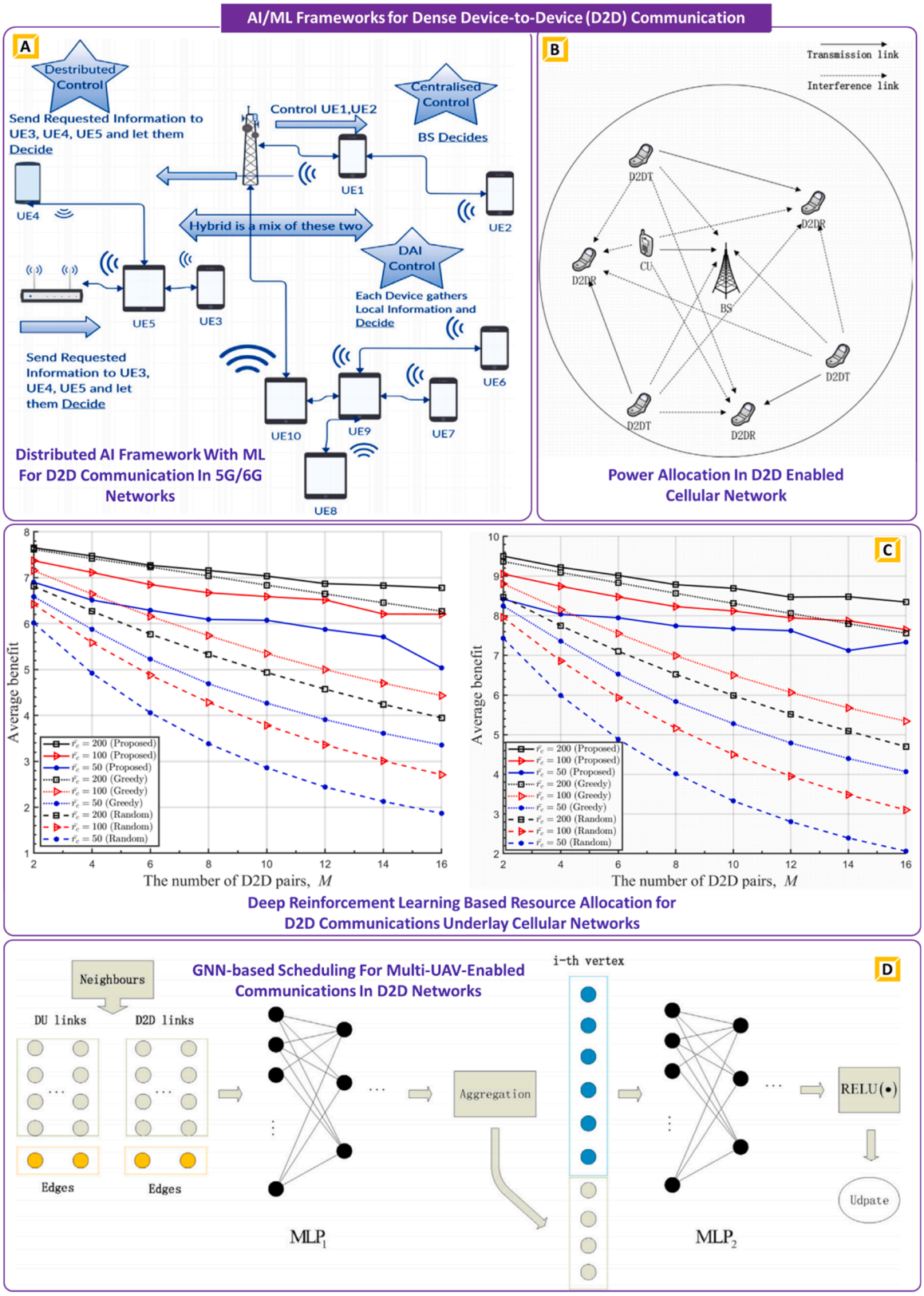


Fig. 5. (A) Control architectures: centralized, distributed, hybrid, and Distributed-AI for D2D resource decisions, taken from [25] with the permission of Elsevier. (B) Underlay D2D with CU resource reuse illustrating co-channel interference and power-allocation challenge, taken from [150] with the permission of Elsevier. (C) DRL-based resource allocation improves average benefit over greedy/random baselines as D2D pairs scale, taken from [152] with the permission of MDPI. (D) GNN-based scheduler pipeline (message aggregation + MLPs) for multi-UAV/D2D link activation, taken from [153] with the permission of Elsevier.

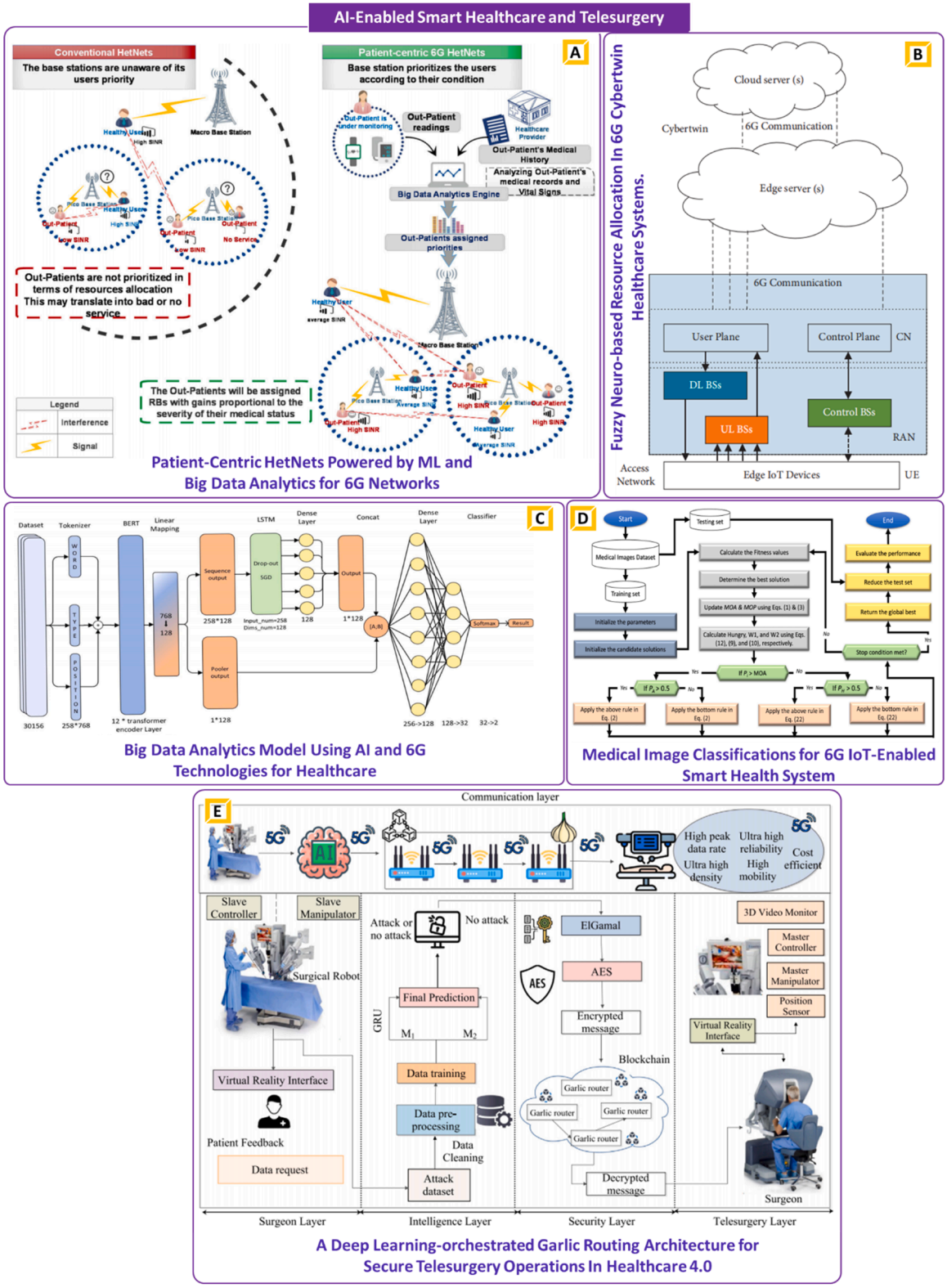


Fig. 6. (A) ML-enabled base stations prioritize users based on health conditions using big data analytics for efficient 6G resource allocation, taken from [159], with the permission of IEEE. (B) An intelligent healthcare architecture leveraging 6G cybertwins and fuzzy neuro models for adaptive, real-time resource management, taken from [160] with the permission of Wiley. (C) A hybrid deep learning pipeline integrating BERT, LSTM, and dense layers for processing healthcare big data in next-gen networks, taken from [161], with the permission of IEEE. (D) Flowchart showing fitness-based feature selection and classification pipeline for medical image analysis in smart health systems, taken from [162], with the permission of MDPI. (E) A layered telesurgery framework combining AI, blockchain, and encryption for attack-resilient, real-time remote surgeries using 6G, taken from [163] with the permission of Elsevier.

5.4. Smart manufacturing and industrial IoT (IIoT)

Smart manufacturing is a fundamental element of Industry 4.0, which encompasses AI/ML, edge computing and the IIoT, thereby aiding predictive quality, adaptive control and operating effectiveness in industry. Manufacturers are thus able to utilize intelligence in their processes from plant and equipment maintenance to the detection of defects, with the result that they are beginning to move from being reactive to proactive decision makers and self-correcting systems. The introduction of IIoT technology will also greatly assist in the interaction of machines, sensors and cloud technology, thus allowing holistic monitoring and control [170–172]. AI models enhance fault detection accuracy, minimize downtime and optimize production workflows. In particular, it is becoming increasingly common to use deep learning and transfer learning solutions to deal with huge sensor datasets, anomaly detection and predictive maintenance in real time, with the potential for significant reductions in operational costs and greater scalability for large corporations and small- and medium-sized enterprises.

Recent case studies exemplify the tangible applications of AI in IIoT-enabled smart factories. In, [173] Abdallah et al. addressed anomaly detection and predictive maintenance using time-series deep learning models (LSTM, AutoEncoder, DeepAR) on multi-sensor manufacturing datasets, shown in Fig. 7 (A). The problem was sparse and heterogeneous sensor data, which was tackled using inter-sensor transfer learning, demonstrating a remarkable 11.6 % accuracy improvement in classifying defect types across MEMS and piezoelectric sensors. This study demonstrated how transfer-learning-enhanced deep models can generalize across multi-source sensor data for reliable industrial fault prediction.

In his subsequent research, Emad H. Abualsauod utilized hybrid pipelines using kernel PCA and GANs to detect faults and showed that AutoEncoder neural networks and One-Class SVMs achieved a high F1-score and precision as far as it is concerned with visual and process anomalies [177]. Abualsauod's contribution highlights the effectiveness of generative–discriminative hybrid architectures in improving anomaly sensitivity and manufacturing resilience. In their work, which studied lightweight real-time solutions for SMEs, Park et al. adopt a YOLOv4 based model and the implementation of computers on a single board with OpenCV to detect liquefied gas defects for light production solutions described in Fig. 7 (B). It results in sustainable, low-cost defect solutions, making AI available for low-technology industrial users [174]. The work demonstrates how edge-deployable AI models can democratize industrial automation in resource-limited settings. Complementarily, Neupane et al. in introduced a 1D CNN model for bearing fault classification, outperforming conventional 2D CNNs in vibration data interpretation from the CWRU dataset [175], shown in Fig. 7 (C). Linking these efforts, demonstrated deep learning's role in early failure identification and emphasized scalable architectures that adapt to variable data quality [178], shown in Fig. 7 (D). At the same time, the work in smart highlighted the importance of federated learning

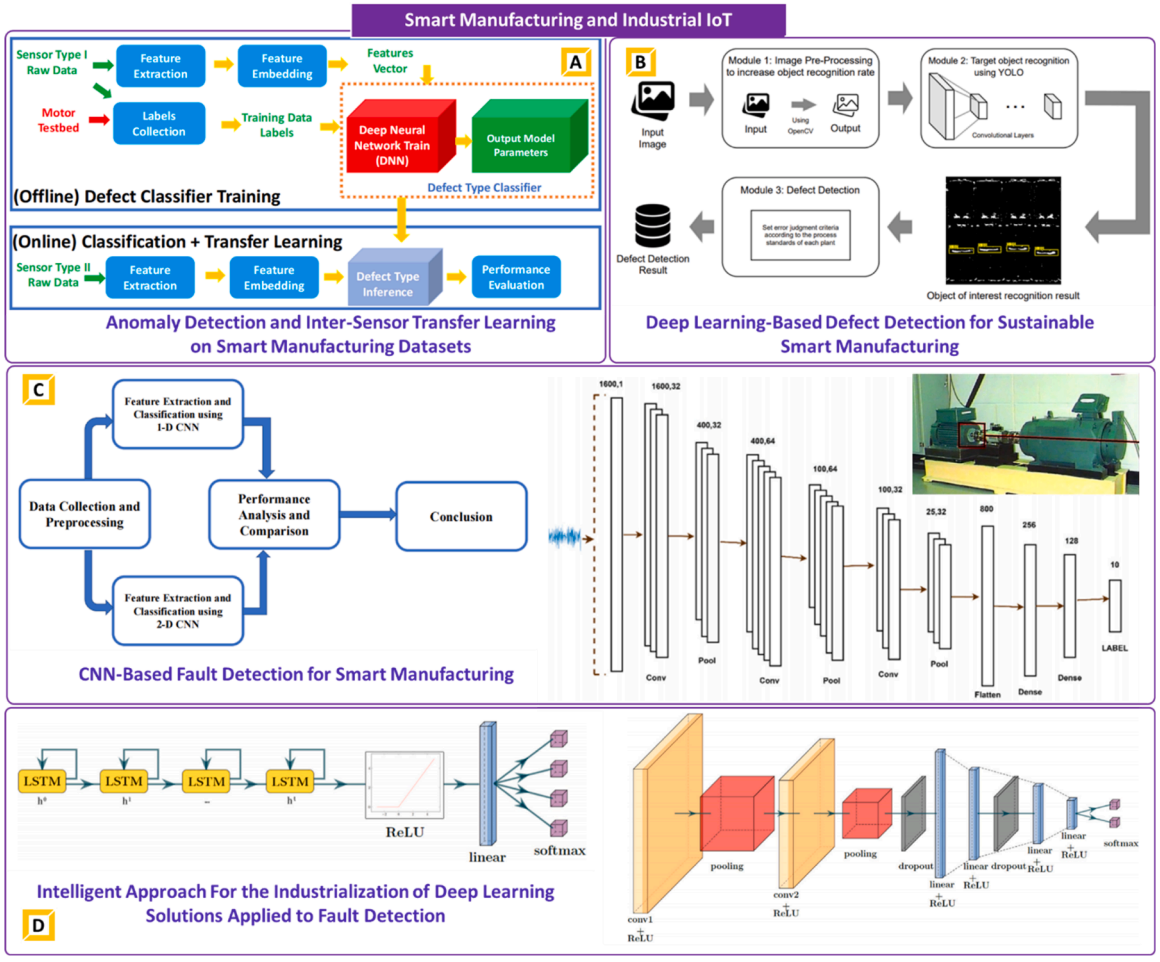


Fig. 7. (A) A DNN-based framework for inter-sensor transfer learning and anomaly detection using offline classifier training and online inference, taken from [173], with the permission of MDPI. (B) A sustainable manufacturing solution integrating image pre-processing, object detection using YOLO, and defect evaluation based on visual standards, taken from [174] with the permission of MDPI. (C) A comparative pipeline using 1D and 2D CNNs for extracting features from industrial signals and classifying faults for smart diagnostics, taken from [175] with the permission of MDPI. (D) Deep learning models, including LSTM and CNN variants applied to vibration/sensor data for accurate industrial fault prediction and system health monitoring, taken from [176] with the permission of Elsevier.

and edge analytics to manage privacy-friendly decentralized industrial environments [176]. Together, these applications paint a picture of a roadmap from inspection with AI-based on centralized computer systems to edge-intelligent, decentralized decision-making systems.

There are still some challenges that remain, despite progress being made. Most models require ample training data, so in the early attempts, it can be slow, especially where unstructured environments require a lot of pre-training. Even though bleeding-edge, high-quality deep learning models would be ideal for anomaly detection, they are largely non-interpretable and that can cause operators to be hesitant about using the more sophisticated approach and can lead to respective regulatory compliance concerns! Interoperability continues to pose an issue, depending on slightly for both the legacy systems and the new AI baked in the IIOT infrastructure. In the near future, topical, applied research needs to be advanced, relative to Interpretable AI models, items pertaining to federated learning, and continual learning, and the establishment of industry standards and protocols for the real-time applicability of IIOT and AI. Modular low-cost AI solutions for SMEs for application in resource constrained areas is also now a major need. Trends indicate a shift toward edge intelligence, real-time co-simulation with digital twins, and collaborative human-AI manufacturing ecosystems that ensure both scalability and sustainability [179–181].

5.5. AI in smart cities

AI is an essential catalyst in the transformation of traditional urban infrastructures into responsive, data-driven smart cities. Using real-time data from IoT devices, edge sensors, and the general public, AI enhances services such as public safety, optimization of energy, traffic control, and environmental monitoring [182–184]. Smart city managers can utilize AI-based intelligent decision making to detect anomalies, predict risks, and dynamically allocate resources. As the smart cities migrate to decentralized collaborative efforts, with 5G/6G connectivity and edge computing enabling ultra-low latency services for disaster prevention, intelligent transportation and crowd-aware surveillance, the use of deep learning and federated learning and explainable AI (XAI) frameworks now envelop the design of the scalable and reliable AI-driven urban systems [185,186].

Multiple case studies illustrate the variety of applications of AI across smart city domains. For example, Reis et al. developed a hybrid federated learning model utilizing AutoEncoders, CNNs, and LSTMs to protect 5G-enabled IoT devices through cyber threats, with an F1-score of

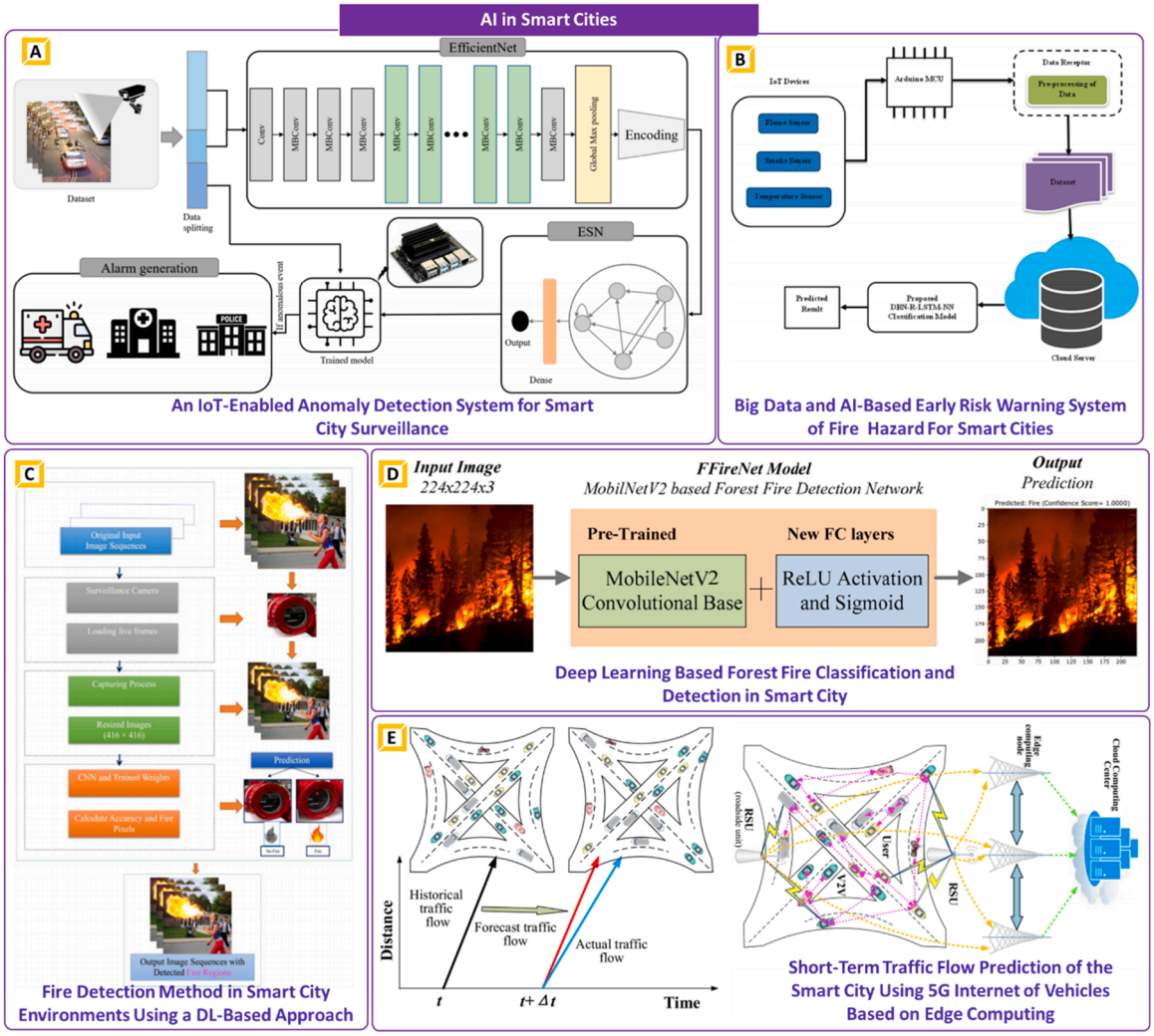


Fig. 8. (A) An EfficientNet-based system using IoT sensors and deep learning for real-time surveillance and emergency alert generation, taken from [188], with the permission of MDPI. (B) Framework combining IoT sensors with deep learning for early fire hazard prediction and cloud-based risk management, taken from [189] with the permission of Elsevier. (C) A CNN-powered model processing image sequences from surveillance cameras to detect and classify fire incidents in real time, taken from [191] with the permission of MDPI. (D) A deep learning model for classifying forest fire images using pre-trained convolutional networks with ReLU and sigmoid activation, taken from [192], with the permission of MDPI. (E) Short-term traffic forecasting in smart cities using Internet of Vehicles, edge computing, and historical traffic data analysis, [193] with the permission of IEEE.

96.8 % [187]. Their work lays the foundation for scalable, privacy-preserving cybersecurity frameworks essential for managing massive IoT networks in 6G-enabled smart cities.

In a similar vein, Islam and colleagues fused a 2D-CNN with Echo State Networks (ESN) for real-time anomaly detection in surveillance videos targeting edge devices with limited computational power, as depicted in Fig. 8 (A) [188]. This approach exemplifies how lightweight AI architectures will support real-time situational awareness and public safety analytics in 6G's edge-intelligent surveillance systems. In [189] Zhang et al. utilized a Deep Belief Network (DBN) with sequential LSTM Neural Networks for the early detection of fire hazards utilizing big IoT data resulting in an accuracy of 98.4 % and negligible errors. In Zhang's study, one can see the synergy made possible due to 6G between IoT sensing and Deep Predictive Analytics for ultra-reliable, mission-critical urban safety applications as set out in the application objectives. Fig. 8 (B) shows the results obtained. In another application, Avazov et al. in [190] enhanced YOLOv4 for real-time fire detection, deploying the model on Banana Pi M3 boards for rapid alert generation in smart city settings. Their edge-optimized framework reflects 6G's vision of ultra-low latency, on-device intelligence for disaster response and early event detection.

To elaborate on this point, in a framework for federated learning, the integrated thermal imaging and sensor data is coupled with Densenet201, and BiLSTM-Dense gives ideal performance (1.0 accuracy, precision, and recall) while keeping the data privacy element. Furthermore, this paper discusses wider smart city processes, adumbrated in other research areas, besides fire and surveillance applications, indicated in Fig. 8 (C) [191]. Khan and Khan [192]. introduced an integrated edge computing framework where environmental monitoring was optimized using a CNN-based AI model on FPGA-enabled devices, ensuring real-time decision-making in air-quality management, shown in Fig. 8 (D). Their research illustrates how 6G-enabled edge AI can decentralize environmental management, ensuring adaptive urban sustainability and low-latency ecological analytics. Finally, Shenghan Zhou et al. discussed a multi-modal sensor fusion approach to optimize traffic flow using GNNs that adaptively modified signal timing based on real-time vehicle density, illustrated in Fig. 8 (E) [193]. Zhou's work captures the essence of 6G's AI-native networking, combining sensing, computation, and communication for intelligent mobility orchestration in smart cities.

In spite of this encouraging backdrop, a number of problems remain. The heterogeneity of data sources, the interoperability of legacy and novel systems, the absence of standardization of AI applications, etc. constitute considerable obstacles. Privacy-protecting learning models such as federate or split learning need stronger validation under potentially adversarial conditions. The ethical issues associated with surveillance and the ownership of data, and with algorithmic bias, have not yet been resolved. Future smart city projects must attach increasing weight to transparency, explainability, and citizen inclusiveness, in AI decision-making. Trends point toward decentralized AI architectures, 6G-integrated sensor ecosystems, and hybrid cloud-edge intelligence as the foundation for building equitable, sustainable, and resilient cities [194–196].

6. Challenges and research gaps

With the advancement of 6G systems towards complete AI-native systems, there are several ongoing and newly emerging challenges which may obstruct their scalability, reliability and retention of societal trust. These problems do not occur in isolation to a specific layer of the communications stack, but rather extend to problems such as data availability, model efficiency, deployment constraints, systems integration, security concerns and regulatory aspects. While AI has shown remarkable potential in improving wireless communications capabilities, from channel estimation to resource allocation, this further complexity introduces a new class of problems, including the need for very large sets of annotated data, real-time decision making with constrained resources, adversarial robustness, and ethical accountability, while the absence of interoperability and standardisation between vendors and platforms is inhibiting the widespread uptake of these innovations and applications. A conceptual visualization of these six foundational challenges, along with their interdependencies, is presented in Fig. 9, offering a root-cause perspective to guide both research and implementation strategies in the AI-enabled 6G era.

6.1. Data scarcity, annotation costs

The performance of wireless systems driven by AI depends critically on the existence of large, high-quality, well-annotated datasets. However, obtaining such datasets is often a formidable task in real 5G or emerging 6G environments. Factors such as high mobility, heterogeneity of hardware, privacy aspects of data, and the costs of data labelling severely limit the scalability of supervised training. Unlike in the domains of computer-vision or NLP, the quantity of data for wireless communications systems is often proprietary, noisy, and scenario dependent. This adds enormous difficulty in generalizing algorithms between environments [32,197].

To mitigate these limitations, a promising avenue is transfer learning in combination with the generation of synthetic data. Transfer learning adapts AI models trained on one domain or network condition by fine-tuning them to operate in a different or data-poor domain, so that the needs of capital and time for the training phase can be drastically reduced. For example, models that experience training in urban cellular topologies can, with limited retraining, be easily expended to rural or indoor scenarios. This scope of reusability improves the robustness of the model with respect to varying environments while lessening the need for costly data collection efforts [198–201].

Alongside transfer learning, GANs and VAEs are increasingly being used to generate realistic synthetic wireless data. These generative models learn statistical features of wireless channels, interference patterns, mobility dynamics to yield synthetic channel impulse responses or CSI datasets which closely mimic realistic operating conditions. Such data augmentation removes sparsity of datasets and also facilitates adversarial robustness testing, emulation of extreme operating regimes and privacy-preserving training by avoiding the need to share raw user-generated databases. For instance, GAN based channel emulators can bootstrap DRL agents for spectrum management in circumstances when real world datasets are limited or expensive to obtain [98–203].

Moreover, the federated learning and split learning paradigms provide data-efficient solutions that allow for collaborative model training instead of data-centric aggregation. Each edge node provides its own local updates resulting from its own small data set, hence improving the global model while preserving the privacy constraints. This distributed paradigm yields not only better annotating costs, but it also captures contextual diversity that exists in multiple deployment environments [204,205]. However, notwithstanding the above advantages, there are a unique set of practical considerations associated with federated learning in wireless networks that should be noted for a balanced perspective, namely: 1) Non IID (non-independence identically distributed) data: Edge devices frequently sense heterogeneous data, being that this is dictated by the varied user mobilities, non-identical hardware devices and channel conditions. Such non-IID distributions lead to slower model convergence times, and reduced global performance compared to the centralized learning paradigms, 2) Communication overhead: FL practices frequent exchanges of nodes characteristics in respect of the model parameters or gradients involved between clients and aggregators. This practice may congest uplink bandwidth that is often of limited availability and referred to as "last-mile bottleneck." This adds latency and energy burden for low powered IoT or mobile nodes [206] 3) Security vulnerabilities: Although FL has potential features for privacy in relation to raw data, it is still allowed to be violated in terms of model-poisoning, gradient-inversion and Byzantine attacks that befall the integrity of the

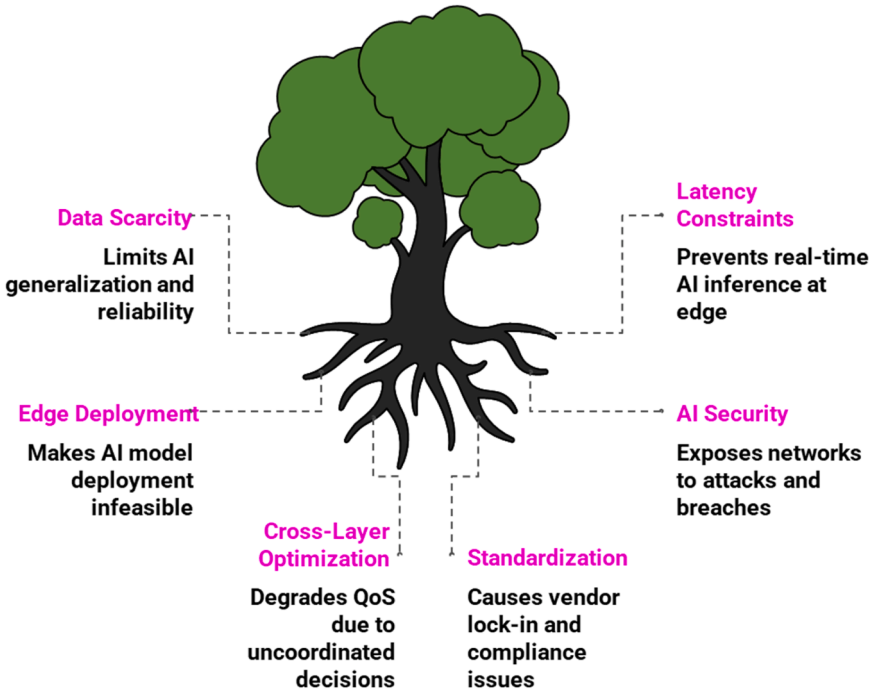


Fig. 9. A root-cause visualization of key AI-related challenges in 6G wireless systems.

model or lead to the leakage of sensitive information [207]. In order to overcome these limitations, recent research has concentrated on following bandwidth-adaptive aggregation, gradient compression and sparsification, asynchronous or clustered updates and secure-aggregation or blockchain-based audit frameworks to foster the implementability and strengthen the robustness, and scalability. However, the eventual, ultimate paradigm of a fully implemented FL yet privacy-improved but communication-efficient in 6G systems will require synergies in relation to the joint optimization of the learning algorithms and the network references so that fairness, resiliency and energy efficiency remains the objective from the outset across the multiple, diverse edge contexts [208,209].

6.2. *Real-Time inference under latency constraints*

The requirement of real-time inference brings another key necessary system requirement imposed by AI based applications in future generations of 6G wireless networks especially in the case of things like the self-driving vehicle mission-critical applications, tele-surgery, industrial control, extended reality and such like applications, all of which require ultra-low latency (sub-millisecond to few milliseconds) reliability of higher degree in order to provide timely, safe and efficient operation [210]. Most of the state-of-the-art AI/ML models for example, like deep neural networks, reinforcement learning AI agents require significant computation effort, which makes them virtually impossible to realise in terms of meeting the strict requirements for latency and energy for real-time applications in practice. Also, real-time inference will typically be located on edge devices or small cell units again having very limited amounts of compute and power available which makes the problem not purely one of algorithmic speed but more of hardware aware design, delay due to preprocessing of data, and unpredictable network jitter which need to be taken into account also to be able to reduce the end-to-end latency requirements [211].

The size and complexity of the AI models are a major source of inference latency. Models such as LSTM, GRU, PPO, and Transformers, although they are effective models, have millions of parameters and involve non-trivial sequential computations, which is not possible to be performed in real times. This is particularly a concern in edge- AI environments where local inference is needed to reduce dependency on cloud processing models, particularly in multi-agent environments, such as vehicular V2X systems, where we need simultaneous model updates, frequent context changes, handovers, and also frequent changes in context (i.e., dynamic changes using model updates, user mobility, etc,), it further exacerbates the timing problem and the constraints involved. Furthermore, the data offloading with the model inference scheduling can also have a significant effect on lateness and throughput, thus contributing to the overall effects of latency and throughput [212,213].

To surmount these challenges, many solutions are under active investigation. Model compression techniques such as pruning, quantization, and knowledge distillation are extensively used to reduce the size of the model and the computation required for it. These methods permit lightweight inference on edge devices such as baseband processors, microcontrollers, or FPGA accelerators [214,215]. The edge intelligence frameworks integrate hardware-aware neural architecture search to optimize the model architecture for latency-sensitive deployment. In addition, split computing and collaborative inference can balance latency and computational load dynamically. These selectively execute partial models on the edge and the rest in the cloud. Recent developments in network slicing and URLLC (Ultra-Reliable Low Latency Communication) in 5G/6G ensure adaptive allocation of network resources for critical inference tasks. Finally, neuromorphic and spiking neural network-based hardware can perform ultra-fast and event-driven computation similar to their biological counterparts, which have the potential to achieve millisecond-level inference. With the advent of 6G, integration of AI Inference Scheduling, Adaptive Model Loading, and Network-Aware Resource Orchestration will be crucial to meet real-time constraints across latency-sensitive wireless applications [216,217].

6.3. *Edge deployment, model compression, and complexity-aware resource allocation*

The 6G network vision characterizes an environment of pervasive intelligence, where models based on artificial intelligence are executed not only in centralized clouds but also in distributed edge environments, base stations, small cells, IoT nodes, and user devices. This new frontier is often called Edge AI, is critical in ultra-low latency decision making

and bandwidth economy. However, the introduction of high-dimensional deep neural network models at the network edge is constrained with regard to memory, processing and power budget capacity limit. The problems are made even bigger across V2X and D2D systems, where mobility, density and real-time safety requirements limit the available latency envelope to a few milliseconds [218,219].

With a view to sustaining responsiveness, modern 5G/6G designs increasingly adopt complexity-aware model engineering, wherein algorithmic precision is traded off against computational and signalling overhead. Model-compression techniques provide a reduction in parameters and arithmetic operations through pruning, quantization, low-rank approximation, knowledge distillation, and tensor decomposition, enabling inference on lightweight processors at minimum loss of accuracy [220]. However, the compression ratio directly impacts latency, inference stability, and communication load during model synchronization. For example, heavy pruning may speed up local inference but lead to more frequent re-training or gradient exchanges, thereby increasing control-plane signalling [221,222].

To resolve this trade-off, researchers are now coupling split computing, where some aspect of a model is executed locally while deeper layers are run inside a cloud, with collaborative inference scheduling that dynamically selects the execution site, depending on instantaneous channel states and queues. Offloading policies built with reinforcement learning and adaptive network slicing use just-enough bandwidth and compute resources on for each inference task, co-optimally using resources in spectrum use and model latency. Such co-design efforts therefore convert the resource allocation problem from a pure radio problem to a joint AI–network orchestration issue [223,224].

Emerging technologies such as hardware-aware neural architecture search, event-driven neuromorphic processors, and spiking neural networks continue to push these boundaries even farther, providing optimisations to both assumptions on both model topology, and to the actual execution paths in the hardware [225,226]. Coupled with federated and split learning, these new approaches provide optimisations to the amounts of raw data to be exchanged, while at the same time keeping the degree of accuracy of models to be bootstrapped, which is particularly important in terms of greater privacy and data-sensitivity metrics of vehicular and IoT application layers [227,228]. Ultimately, the success of scalable AI deployment will be dependent on quantized growth either required of the models, or the number of hidden layers or weights at hand in such networks in dense ecosystems such as 6G networks ever-growing will depend upon, of greater importance here, are standardization of metrics of complexity (i.e. FLOPS-per-decision, signalling cost-per-update) of self-adaptive inference controllers (generally accepted-level bandwidth) and the models' degree of weights, or hidden layers, or hidden nodes determinable depending on bad radio and CPU resources combined. In fact, in future, edge-deployment in 6G must become a greater framework of complexity-performance co-optimisation, so that each marginally greater bit of data-intelligence yields adequate network value in terms of performance resultant (under--traffic) of adequately bounded energy levels accrued whilst still avoiding undue latency [61,229].

6.4. AI security and adversarial robustness

In the context of 6G wireless, AI will permeate the control, management, and data planes deeply enough that security and robustness of learning models will no longer be optional but mission-critical. The open, heterogeneous, and distributed nature of next-generation wireless systems exposes AI and ML components to a wide range of threats, including data poisoning, model inversion, backdoor, and evasion attacks [230,231]. Particularly in safety-critical domains such as autonomous vehicles, telesurgery, military communication, and industrial automation, those threats are very serious, as compromised AI decisions may directly cause physical damage or system failure. Unlike deterministic rule-based algorithms, data-driven models can be easily fooled by imperceptible perturbations in their input data into generating incorrect yet seemingly legitimate outcomes.

Today, protecting AI/ML systems far goes beyond intrusion detection. All steps of the pipeline-from collecting data to training, combining, and deploying models-can be attacked. Because modern networks use continuously trained models in many tasks, such as traffic prediction and spectrum management, a single attack can proliferate throughout the system [230]. Threats include poisoning the data with bad data, poisoning the model with tampered training updates, and adversarial attacks that steal or expose private information. Backdoors can also be introduced by hackers. Attackers in wireless communication can generate fake signals or patterns to mislead AI systems that govern tasks such as modulation, channel estimation, or resource scheduling [232]. This even goes down to small changes, like changing pilot signals, which may cause the device to choose the wrong beam or trigger false alarms on vehicle-to-vehicle links, putting the reliability and safety of 6G networks at risk. In this regard, the defensive strategy requires a multi-layer approach: adversarial training strengthens models by exposing them to normal and tampered data, while robust aggregation in federated learning forms the second line of defence to block poisoned updates. Privacy tools, including differential privacy and secure encryption, protect sensitive data, and the use of blockchain frameworks or zero-trust architecture ensures updates cannot be secretly altered. Finally, XAI allows humans to identify unusual model behaviour and provides an additional layer of protection against threats [233].

6.5. Cross-layer and cross-domain optimization challenges

The aim of the vision in 6G is to consider AI as a strong tool that manages and further enhances each component of the communication system, starting from physical signals and access control to the network, transport, and application layer. Balancing tough and sometimes conflicting demands-such as very low delay, extremely high data speeds, strong reliability, better security, and efficient energy use-can be achieved by AI through its coordination across all these layers. In simpler terms, AI will make 6G networks run smarter and smoother to cater to the most demanding requirements of the future [234,235]. However, most existing AI models are designed in layer-specific silos, optimizing individual tasks like beamforming, handover, or routing without accounting for upstream or downstream dependencies. Such isolated optimization can result in suboptimal global performance, network instability, or unintended trade-offs, especially under dynamic user mobility, multi-service traffic, or coexisting network technologies (e.g., 5G NR + WiFi + THz links).

In 6G, communication, computing, control, and sensing fields all converge while at the same time presenting new problems in having them all performed effectively at once. For example, autonomous vehicles depend upon AI to link up the vehicle-to-everything links, edge computing and traffic prediction, while telesurgery rests upon smooth coordination between network quality, image processing, robotic control and cloud-enabled AI. Each domain has its own rules to respect and its needs, so for the overall best performance from the supporting technology there needs to be joint learning models to cross domains, alter to time and space changes, and see which goals are best learned simultaneously.

There are algorithmic and architectural innovations that will be needed to address such challenges. Multi-agent reinforcement learning is a particularly exciting paradigm to find ways in which decentralized agents can coordinate across layers and domains through collaborative policy learning [236,237]. Graph Neural Networks are well-suited to model complex, non-Euclidean relationships between network elements and can support topology-aware, multi-layer optimization. Furthermore, forms of hierarchical AI architectures are being designed, in which high-level controllers prescribe policy to distributed task-specific AI modules, lending coherency to stack policy. With regard to

architecture, digital twin platforms can be employed in order to simulate and train cross-layer policies before real-world deployment, which will lower risk and drive convergence. Lastly, semantic-aware AI models are emerging, which have a focus on the meaning or value of data (rather than raw bits), allowing for more efficient and context-related optimization [238,239].

6.6. Standardization, interoperability, and ethical concerns

Challenges related to standards, interoperability, and ethics become paramount when 6G is moving toward AI-driven networks. Unlike previous systems with clear rules such as 3GPP or IEEE, there are no common protocols for the design, testing, and deployment of AI models. The integration is tough to achieve due to various frameworks such as PyTorch, TensorFlow, and ONNX on multiple platforms: cloud, edge, and devices. In practical 6G scenarios with many vendors, operators, and regulators, ensuring AI systems can work smoothly together remains a big challenge. Without shared standards, interfaces, and benchmarks, it will be hard to ensure reliability, scalability, and trust across organizations [240].

In parallel, Ethical issues will also grow in importance, considering that AI is becoming an integral part of 6G networks. Since AI will be responsible for the control of basic functionalities such as traffic, network slicing, emergency handovers, and security, major concerns will revolve around bias, fairness, transparency, and accountability. Biased training data will lead to unfair resource allocation, while black-box models used for telesurgeries and self-driving are not appropriate to provide explainability and, more broadly, legal responsibility. Privacy risks are greater since enormous amounts of real-time user data are collected and analyzed; in fact, even state-of-the-art privacy tools like federated learning or differential privacy require careful balancing between accuracy and protection. Lastly, compliance and policy alignment are further complicated due to various data protection laws around the world, such as GDPR and CCPA [240].

In fact, global efforts are underway to create common standards for the use of AI in telecom as 6G develops. The groups currently working on rules for explainable AI, model management, certification, and risk control include ITU-T, ETSI, 3GPP, and IEEE. Proposals include AI registries, zero-trust security, and governance tools to make sure transparency and accountability are maintained. On the technical end, there will be standards needed regarding model format, updates, and monitoring so that systems can work seamlessly across different platforms. Practices such as keeping humans in the loop, checking for fairness, and performing impact assessments would help social values and laws to be consistent in terms of ethics. And this is going to require collaboration by engineers and policymakers with ethicists and society to build responsible interoperable AI for 6G [241,242].

6.7. Model-specific challenges and design trade-offs in AI-driven 6G networks

The advent of AI has brought substantial advances in predictive modelling, control optimization, adaptive resource allocation for 5G and 6G networks, but it will carry its own design trade-offs and deployment constraints for each model architecture [224]. It is, therefore, rather complex to select suitable models (CNNs, RNNs, LSTMs, GRUs, RL agents and GNNs as well as their different architectures) that are suitable for the different temporal, spatial and contextual characteristics of the particular task, hardware and latency limits imposed on edge and base-station nodes, which are involved. Grasping these various model based limitations is important in order to gain the necessary optimisations needed for the devices used to meet the respective stringent ultra-reliable low-latency communication (URLLC) and high-reliability low-latency communication (HRLLC) requirements connected with 6G [243,244].

Among the sequence models, the GRU provides an effective trade-off between performance and computational complexity. It simplifies the LSTM gating mechanism by only using a single update gate by merging the input and forget gates, and removing the cell state. This results in about 25 % fewer parameters, making GRUs more favorable in edge deployment and real-time inference scenarios. The parameter complexity can be mathematically expressed as: $\theta_{GRU} = 3\,(nh + h^2 + h)$ compared to LSTM's $\theta_{LSTM} = 4\,(nh + h^2 + h)$ where nis the input dimension and hthe hidden dimension. This reduction in parameters directly lowers FLOPs, latency, and memory consumption, leading to faster training and inference in delay-sensitive tasks such as channel prediction, mobility forecasting, and dynamic resource allocation [245,246].

However, these advantages of GRU come with challenges that are not trivial. First, although it better handles the vanishing gradients compared to vanilla RNNs, it is still not able to manage extremely long temporal dependencies so common in non-stationary wireless environments. Second, GRU-based systems remain highly sensitive to hidden dimension selection. A smaller hidden size $(h < 64)$ limits representational capacity, degrading forecasting accuracy, while excessively large $(h < 128)$ increases latency, overfitting risk, and energy consumption. Experimental ablation studies from multiple 5G edge-AI deployments indicate that accuracy saturates between hidden sizes of 128–256, whereas computational cost continues to rise almost quadratically with h. Hence, an optimal configuration is typically found at $h = 64 - 128$ with one to two GRU layers and a dropout rate between 0.1–0.3 [247,248].

An ablation protocol was developed to demonstrate this trade-off, testing configurations of hidden size, number of layers and dropout probability. The results demonstrated that increased hidden size improved accuracy to an optimum with detrimental increases to latency per inference beyond this point, which is not compatible to URLLC-dependent tasks. Similarly deep architectures having three or more layers had diminishing returns for their improved accuracy, which was coupled with greater energy usage and inference latency. This means that GRUs provide the best compromise between accuracy and model performance for mid-range edge devices such as UAV controllers or baseband processors, which require moderate temporal modelling performance but critical latency in inference times.

Aside from the GRU, similar constraints limit other AI architectures as follows. CNNs and DNNs give good spatial representation but are seriously compute intensive and sensitive to dataset shift. RL models require large interaction data to be fed into them, are sample inefficient and prone to instability in dynamically changing environments where wireless channel conditions alter rapidly. GNNs, although they model well inter-node relationships in D2D or V2X situations, have a scalability and communication problem when their graph incrementally increases. GANs and VAEs are effective in overcoming problems of lack of data through synthetic generation of channels, but they have issues of instability in training, mode collapse, realistic modelling physical structures. The comparative disadvantages and deployment trade-offs of the major AI models including GRU, LSTM, CNN, RL, GNN and Generativs are shown in Table 8, which gives a short cross- comparative model view of computational complexity, real-time application appropriateness and method of overcoming their short-falls in 6G deployment and usage.

7. Conclusion and future direction

In this review, it has been highlighted that AI and ML are not just supporting constituents, but are the fundamental enablers of the emerging 6G paradigm. The technologies within AI, through deep learning, reinforcement learning, generative and graph models have been seen to plug into the major wireless functions such as beamforming, channel estimation, resource allocation and anomaly detection. The gestating impact of AI, through cross-layer working and real-world application, through a plethora of use cases such as autonomous

Table 8
Model-specific challenges and design trade-offs for AI in 5G/6G [95,249–253].

Model / Technique	Why/When It's Used (5G→6G)	Core Challenges in Practice	Impact on Deployment (Edge/ Cloud)	Practical Mitigations / Notes
GRU (RNN family)	Temporal dynamics (traffic, mobility, CSI) with tighter edge budgets than LSTM	Still sequential (latency); sensitive to hidden size h; struggles with very long dependencies	Good fit for edge/hybrid when tuned; lower params than LSTM reduce latency/energy	Start h=64–128, 1–2 layers, dropout 0.1–0.3; monitor Accuracy/ms & Accuracy/J; quantify gains vs. LSTM.
LSTM (RNN family)	Longer temporal memory than GRU for difficult sequences	Highest param/compute among RNNs; higher inference latency	Often cloud/edge-assisted; may violate URLLC on-device	Use pruning/quantization; offload via split computing; consider GRU replacement.
CNN/DNN	Spatial features (beamforming, modulation recognition)	Compute/memory heavy; dataset shift; explainability gaps	Best centralized/cloud; edge possible only with compression	Apply model compression, edge accelerators, and XAI interpretability for reliability.
RL / DRL / MARL	Adaptive control (spectrum, slicing, power)	Sample inefficiency; instability; reward shaping; partial observability	Distributed/Hybrid deployment; control-plane overhead; convergence risks	Use safety constraints, federated MARL, and lightweight policies.
GNN	Graph-structured radio topologies (inter-cell, D2D, V2X)	Message-passing cost; oversmoothing; scaling on dynamic graphs	Edge controllers/distributed; communication overhead between nodes	Apply sparse graphs, neighborhood sampling, and efficient batching.
GAN / VAE (Generative)	Synthetic channels/CSI; augmentation; robustness testing	Training instability (GAN), realism gap; compute overhead	Usually cloud/offline training; deploy outputs, not models	Add physics-informed priors, validate against real-world data.
Federated Learning	Privacy-preserving collaborative training (vehicular/health IoT)	Non-IID data; uplink overhead; poisoning risks	Federated edge + cloud; slower convergence without care	Use gradient compression, secure aggregation, and clustered updates.
Split Learning	Edge-cloud co-training with reduced client load	Split-point selection; privacy vs. latency; orchestration complexity	Hybrid; depends on backhaul quality	Adaptive splitting; QoS-based slicing; secure intermediate layers.

driving, telesurgery, industrial IoT and Smart Cities, has created intelligent and efficient wireless ecosystems that are both resilient and adaptable. A key point in the paper was the reiteration of the need to outline performance metrics of AI functions and deployment norms, to enable scalable, trusted and efficient AI-powered wireless systems. In looking forward to challenges in the research need to address the following: The self-supervised learning and reduction of supervised data annotation requirements; ultra-low latency AI inference through model compression and edge AI designs; AI robustness to adversarial attacks. Cross-layer and cross-domain optimization should be seamless, necessitating a collaborative approach to AI frameworks and digital twins. There must be a parallel evolution of standardization, interoperability and governance of ethical AI to assist with the fair and equitable deployment of 6G. Some promising research areas are: Hybrid physics-AI models, Multi-agent reinforcement learning, Quantum AI synergy and Federated architectures with privacy-preserving features. These will be key to building AI native 6G networks, which are intelligent, but also safe, explainable, and human-centred.

Availability of data and material

Data will be made available based on a reasonable request.

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CRediT authorship contribution statement

Dhiraj P. Tulaskar: Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Battina Sindhu:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis. **Nitin Chakole:** Writing – review & editing, Resources, Methodology, Formal analysis, Data curation. **Rina Parteki:** Writing – review & editing, Software, Resources, Project administration, Methodology. **A. Anny Leema:** Writing – review & editing, Resources, Project administration, Methodology. **P. Balakrishnan:** Writing – review & editing, Validation, Supervision, Resources, Investigation, Formal analysis. **Ankita Avthanka:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Rangnath Girhe:** Writing – review & editing, Resources, Project administration, Methodology, Investigation. **Madhusudan B. Kulkarni:** Writing – review & editing, Visualization, Validation, Supervision. **Manish Bhaiyya:** Resources, Software, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they do not have any conflict of interest in this proposed work.

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